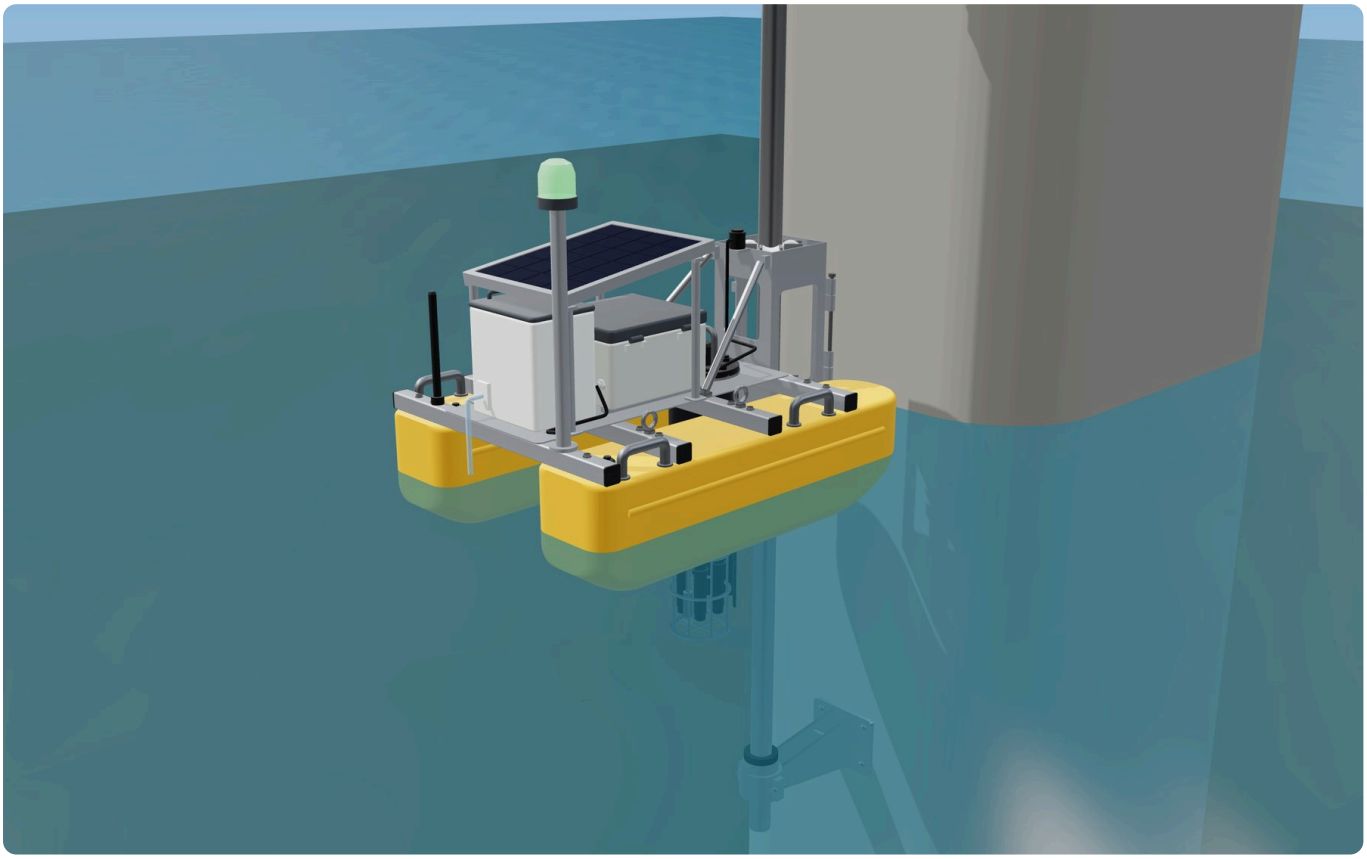
AAKRUTI Innovation Competition 2026 · India Semi-Final · Robotics · AG26-1500$41

AquaSentinel R3

Complete project dossier: design, engineering, electronics, costs, business case, prior art, pitch script and jury answers



A river robot on a bridge-pier guide pole, designed to read the river every 15 minutes, decide on board, and when a spill passes, turn its beacon red, send an alert and bottle a sample for the lab.

Team Sanjivani Hero's Journey · Sanjivani University, Kopargaon · Soham Kadu (Project Lead), Shriraj Kamble (System Design), Bhakti Kadam (Electronics Lead), Tanmay Pendbhaje (Software Lead) · Mentor: Dr. Kiran Wakchaure

Version of 24 Oct 2026. Every number in this document is traced to a project file or a cited source. The tags below say how each claim is supported; nothing here has been tested in a river yet.

Designed Calculated Simulated Estimate / assumption Published fact Not yet tested

Contents

1. Executive summary
2. The problem
3. The solution and how it works
4. Design: the R3 assembly
5. Engineering checks and simulation
6. Electronics, power and firmware
7. Detection logic and analytics
8. Cost: the costed bill of materials
9. Business model, market and the ask
10. Prior art and competition
11. Status, roadmap and risks
12. Engineering opportunities (proposals)
13. The 8-minute pitch script
14. Jury questions and answers
15. Gaps, missing links and claims to verify
16. Project folder map and files
17. Sources

1. Executive summary

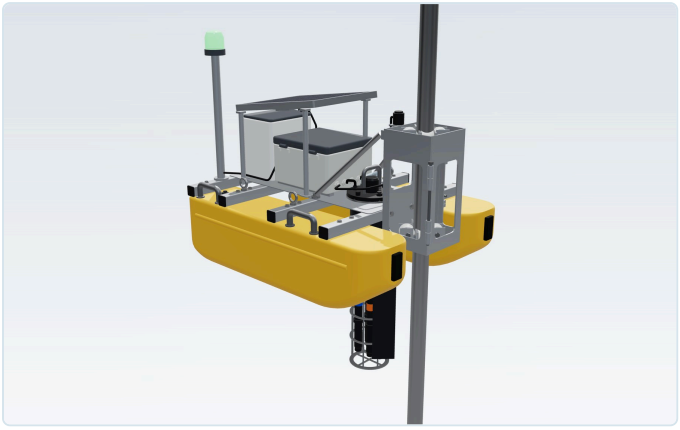
Topic	What we can say	Support
Problem	India: 296 polluted river stretches/locations; Maharashtra 54, the most (CPCB, published 2025). Most monitoring is manual, monthly or quarterly.	Published fact
Product	Float on a 2-inch 316L guide pole clamped to a bridge pier; senses pH, EC, DO, turbidity, temperature, level every 15 min; decides on an ESP32; acts with a beacon, LTE/SMS alert and a 4-bottle sampler.	Designed
Mechanics	Mass 48.54 kg (CAD estimate); freeboard 143 mm; reserve buoyancy 2.27×; righting arm positive to 60°; rides 2 m of level on 8 UHMW rollers; anti-jam margin 8.62.	Calculated
Structure	Pole safety factor 2.66 in our most conservative static load case (1.5 m/s current + 500 N debris); FEA with fixed ends 4.4. Dynamic impact, fatigue and anchors not yet verified.	Calculated Simulated
Power	≈ 4.11 Wh/day with an assumed beacon load; 61.4 Wh usable; ≈ 12 days without sun (preliminary).	Estimate / assumption
Detection	4 of 4 injected spills flagged within 30 min on 30 days of synthetic data. Proves the logic, not field accuracy.	Simulated
Cost	Prototype float ₹157,492 + bridge kit ₹29,218 = ₹186,710 ex-GST, all 66 part numbers priced from Indian listings (23 Oct 2026). At 100 units ₹148,055.	Estimate / assumption
Business	Monitoring as a service: ₹12,000/month per float, 5-year contract; hardware + installation paid back in ≈ 20 months; ≈ 46 % margin over the contract.	Estimate / assumption
Status	Designed, calculated, simulated. Not built full size; not tested in water. Next: 1:5 model in water and electronics bench demo before the 19–20 Nov finale.	Not yet tested

2. The problem

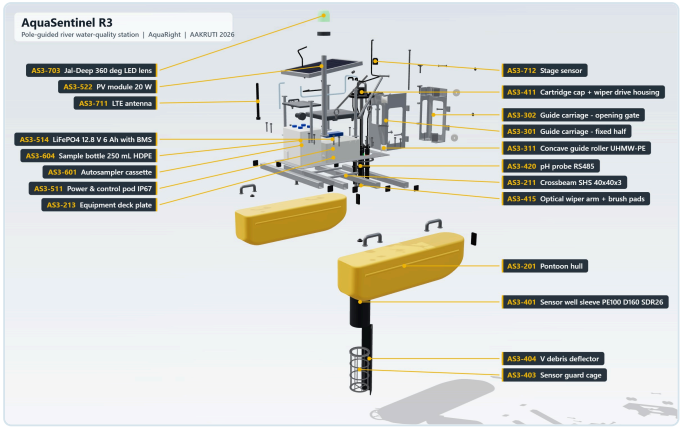
Fact	Value	Source
Polluted river stretches/locations, India	296 on 271 rivers (156 stretches + 140 locations), assessment published 2025 on 2022–23 data; earlier 2022 count 311	PIB 16 Mar 2026; CPCB; Lok Sabha AU2546
Maharashtra	54, the most of any state (55 in 2022)	PIB; CPCB
National monitoring network	≈ 4,922 locations, 2,265 on rivers; manual samples, monthly or quarterly	CPCB reply to NGT, OA 413/2025
Real-time station cost	₹35–40 lakh for 10-parameter instruments; ≈ ₹50–84 lakh per station with systems	CSE; Lok Sabha reply 5115
Delhi tender	32 real-time stations for ≈ ₹22 crore (≈ ₹69 lakh each), later 41 with 5 years of O&M	The Week, May 2025

A discharge that lasts a few hours, often at night, falls between scheduled samples. Real-time stations exist but cost tens of lakhs each, so they are sparse. AquaSentinel is designed as a dense, low-cost early-warning layer between them, not a replacement.

3. The solution and how it works



Concept render of the R3 design (from our SOLIDWORKS CAD)



Exploded view, labelled with part numbers

Function	How (design)	Status
SENSE	pH, EC (K = 10), optical DO, turbidity 0–1000 NTU (DFRobot RS485 Modbus probes, IP68) on a pull-up cartridge; water temperature; river level by an upward ultrasonic sensor	Designed
THINK	ESP32-WROOM-32E reads every 15 min and decides on board: hard limits act at once; otherwise a per-hour-of-day baseline (EWMA) with a robust z-score > 3; EVENT when 2 of the last 3 readings are unusual; normal after 4 clean readings; first 7 days learning	Designed Simulated
ACT	360° beacon turns red; LTE Cat-1 alert (SMS fallback); peristaltic pump fills one 250 mL bottle per event (4 bottles, insulated carousel)	Designed
MOVE	No drive motor: the river lifts the float along the pole through 2 m; 8 concave UHMW-PE rollers in two tiers 350 mm apart; side area aft of the pole so it trails the current like a weathervane	Calculated
POWER	20 W solar canopy, 12.8 V 6 Ah LiFePO4, CN3791-class MPPT	Estimate / assumption
SERVICE	Tool-free carriage gate; cartridge lifts out through the deck; wiper cleans the optical windows every 6 h; monthly service visit	Designed

Operating alarm limits (ours, tunable per site): pH below 6 or above 9; DO below 4 mg/L (the CPCB Class C/D criterion); EC above 2,250 µS/cm (the CPCB Class E limit). These are operating limits, not one CPCB class. AquaSentinel is an early-warning system, not a chemical identifier: oil that changes none of its readings can be missed; the lab sample confirms what a spill was.

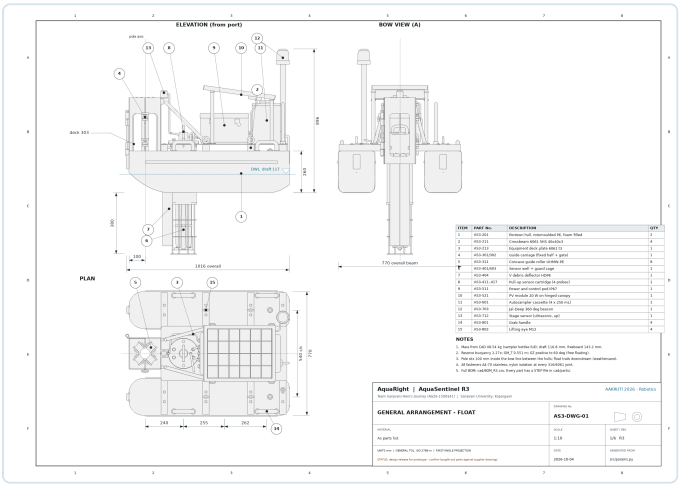
4. Design: the R3 assembly

Designed in SOLIDWORKS as a 66-part assembly (Pack and Go: 66 part files + 1 assembly = 67 files), with STEP, STL and GLB exports, a 6-sheet A3 drawing set and a generated design-check sheet. Masses are partly catalogue estimates; weigh the parts before the tank test. **Designed**

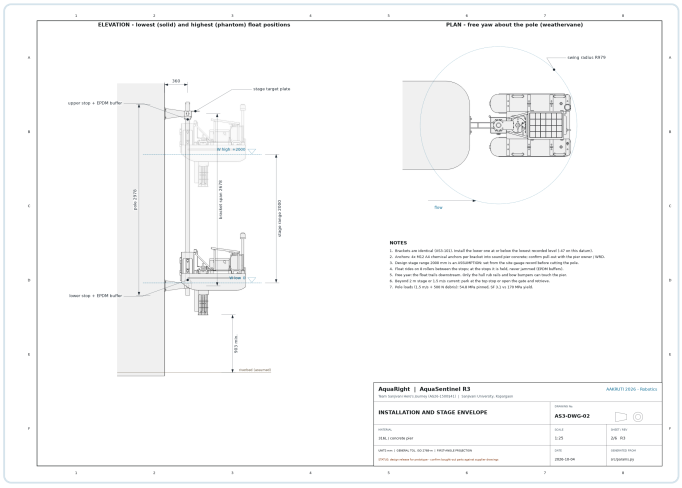
4.1 Key specifications

Item	Design value
Mass (CAD estimate)	48.54 kg; 43.6 kg to carry
Draft / freeboard	117 mm / 143 mm (requirement ≥ 100 mm)
Hull volume / displaced	110.3 L / 48.54 L → reserve buoyancy 2.27×
Stability	GM_T 0.551 m, GM_L 2.19 m; GZ max 180 mm at 25°; deck edge immersion 20.4°
Design current, drag	1.5 m/s; Cd 1.05, frontal area 0.1202 m² → 142 N; + 500 N debris
Guide pole	2 in Sch 40S SS316L, 2978 mm long, bracket span 2678 mm
Carriage	8 rollers, tiers 350 mm apart; anti-jam margin 8.62 (μ 0.05), 1.72 with seized rollers (μ 0.25)
Parts	66 part numbers, 144 pieces: {'part': 34, 'purchased': 20, 'fastener': 9, 'harness': 3}

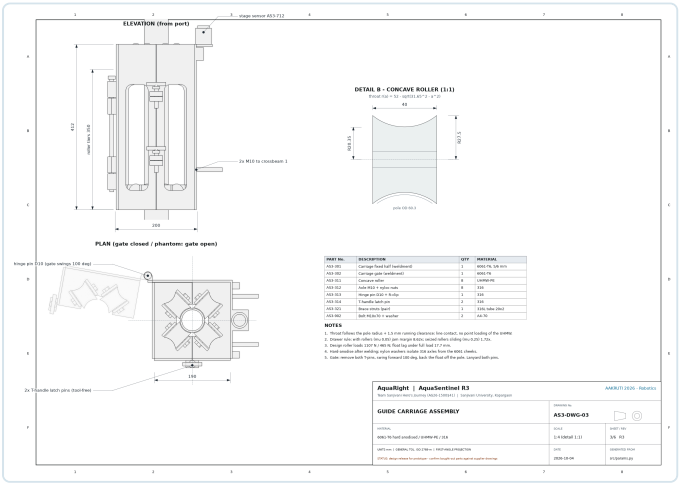
4.2 Drawings



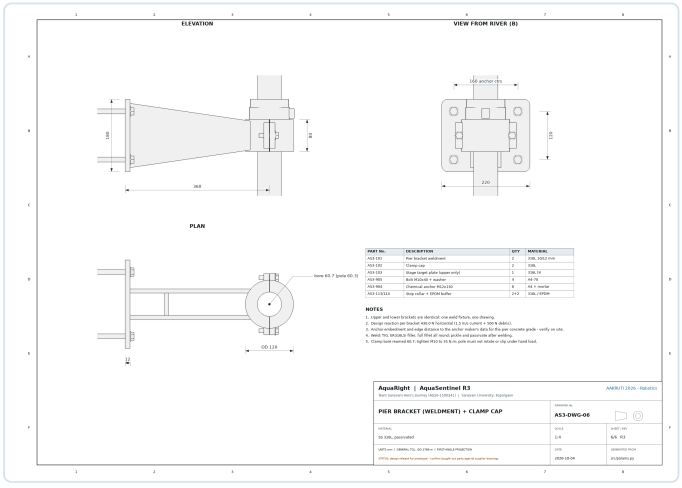
DWG-01 General arrangement



DWG-02 Installation and stage envelope



DWG-03 Guide carriage



DWG-06 Pier bracket

4.3 Complete bill of materials (BOM_R3.csv, all 66 part numbers)

Part	Description	Qty	Material	Process	kg each
AS3-101	Pier bracket (weldment) - upper and lower identical	2	Stainless steel 316L	Laser-cut 316L 10/12 mm, TIG welded, passivated; 4x M12 chemical anchors	12.698
AS3-102	Pole clamp cap	2	Stainless steel 316L	Machined 316L	2.941
AS3-103	Stage target plate	1	Stainless steel 316L	4 mm 316L welded to the upper clamp cap	0.456
AS3-111	Guide pole 2in Sch 40S SS316L	1	Stainless steel 316L	ASTM A312 TP316L pipe, cut to length, ends deburred	16.414
AS3-112	Pole top cap	1	Acetal (POM) black		0.124
AS3-113	Stop collar (split, clamp-on)	2	Stainless steel 316L	Machined 316L, M8 clamp screw	1.245
AS3-114	Stop buffer ring (EPDM)	2	EPDM	Moulded EPDM 70 ShA	0.099
AS3-201	Pontoon hull (rotomoulded, foam filled)	2	Rotomoulded PE, RAL 1003 signal yellow	Rotomoulded PE, one mould for both hulls; PU foam fill; moulded-in M8 inserts	6.821
AS3-202	Bow bumper pad	2	EPDM	Moulded EPDM, bonded	0.075
AS3-203	Hull vent / test plug	2	Acetal (POM) black	Purchased PE threaded plug	0.012
AS3-211	Crossbeam SHS 40x40x3	4	Aluminium 6061-T6, anodised	Saw-cut 6061-T6 SHS, drilled for M8	0.871
AS3-212	Beam end cap	8	Acetal (POM) black	Purchased PE plug	0.009
AS3-213	Equipment deck plate	1	Aluminium 6061-T6, anodised	Laser-cut 3 mm 6061 tread plate	1.99
AS3-301	Guide carriage - fixed half (weldment)	1	Aluminium 6061-T6, anodised	Laser-cut 6061 plate 5/6 mm, TIG welded, hard anodised; nylon isolation at 316 axles	2.806
AS3-302	Guide carriage - opening gate (weldment)	1	Aluminium 6061-T6, anodised	Laser-cut 6061 plate, TIG welded, hard anodised	1.914
AS3-311	Concave guide roller UHMW-PE	8	UHMW-PE	CNC-turned UHMW-PE; throat profile = pole radius + 1.5 mm	0.057
AS3-312	Roller axle M10 + nyloc nuts	8	Stainless steel 316 (brushed)	316 shoulder bolt, nylon isolation washers	0.065
AS3-313	Gate hinge pin D10	1	Stainless steel 316 (brushed)	316 bar + R-clip	0.189
AS3-314	T-handle latch pin (tool-free)	2	Stainless steel 316 (brushed)	316 bar + welded T-bar, lanyard eye	0.072
AS3-321	Carriage brace struts (pair)	1	Stainless steel 316L	316L tube 20 x 2, flattened ends, bolted through deck into crossbeam 2	0.85
AS3-401	Sensor well sleeve PE100 D160 SDR26	1	PE100 black	Cut PE100 pressure pipe	1.053
AS3-402	Hatch flange ring (quarter-turn seat)	1	Stainless steel 316L	Laser-cut 6 mm 316L ring, 6x M6 into deck	0.394
AS3-403	Sensor guard cage	1	Stainless steel 316L	316L rod D6 + laser-cut rings, TIG welded; screwed to the sleeve	1.82
AS3-404	V debris deflector	1	PE100 black	8 mm HDPE sheet, heat-bent V, bolted under crossbeam 1	0.718
AS3-411	Cartridge cap + wiper drive housing	1	Acetal (POM) black	Machined acetal; IP68 servo housing; 4x M16 cable glands	0.752
AS3-412	Cartridge lift handle	1	Stainless steel 316 (brushed)	316 tube 16 x 1.5 bent	0.15
AS3-413	Cartridge spine + wiper shaft	1	Stainless steel 316L	316L tube 25 x 1.5 + 316 shaft D8	0.837
AS3-414	Probe holder spiders (x2)	1	Acetal (POM) black	Machined acetal, split collars	0.117
AS3-415	Optical wiper arm + brush pads	1	Stainless steel 316L	316L flat bar 14 x 5 + nylon brush pads	0.123

Part	Description	Qty	Material	Process	kg each
AS3-416	Probe cables (in-well run)	1	Marine cable	Supplied with probes (mass counted with the probes)	0.0
AS3-417	Sampler intake strainer + riser	1	Stainless steel 316 (brushed)	316 mesh strainer + 10 mm tube + quick-connect	0.15
AS3-420	Probe body (envelope)	4	Acetal (POM) black	Purchased; D32 x 300 envelope - replace with the supplier STEP	0.45
AS3-421	ID band - pH probe	1	pH ID band (blue)	Colour-keyed PE sleeve	0.005
AS3-422	ID band - EC probe	1	EC ID band (orange)	Colour-keyed PE sleeve	0.005
AS3-423	ID band - Optical DO	1	DO ID band (green)	Colour-keyed PE sleeve	0.005
AS3-424	ID band - Turbidity probe	1	Turbidity ID band (magenta)	Colour-keyed PE sleeve	0.005
AS3-501	Pod mounting rails (pair)	1	Aluminium 6061-T6, anodised	6061 angle 25 x 25 x 3	0.451
AS3-511	Power & control pod IP67 (body)	1	Polycarbonate RAL 7035	Purchased PC enclosure, machined for glands, pressure-equalising vent	1.433
AS3-512	Pod lid (hinged)	1	Polycarbonate dark grey	Supplied with the enclosure; EPDM gasket	0.798
AS3-513	Pod mounting plate	1	Aluminium 6061-T6, anodised	2 mm 5052 plate	0.358
AS3-514	LiFePO4 12.8 V 6 Ah with BMS	1	LiFePO4 pack		0.8
AS3-515	Controller board (ESP32, RS485, RTC, microSD, LTE Cat-1)	1	PCB		0.2
AS3-516	MPPT charge controller	1	Polycarbonate dark grey		0.15
AS3-521	PV module 20 W (frame + backsheet)	1	Aluminium 6061-T6, anodised		1.55
AS3-522	PV module 20 W (cells + glass)	1	Mono-Si PV cells + glass		0.45
AS3-523	Canopy stanchions + hinge / pin blocks	1	Aluminium 6061-T6, anodised	6061 tube 20 x 2, welded base plates; aft pair hinged, front pair quick-release pins	0.512
AS3-601	Autosampler cassette (insulated body)	1	PE white	Twin-wall rotomoulded PE, PU insulated	1.904
AS3-602	Autosampler lid (hinged, tamper-seal hasp)	1	Polycarbonate dark grey	Moulded PC, EPDM gasket	0.445
AS3-603	Bottle carousel (lift-out cassette)	1	Acetal (POM) black	Machined acetal; the whole carousel swaps out (chain of custody)	0.199
AS3-604	Sample bottle 250 mL HDPE (lab-supplied)	4	PE white		0.29
AS3-605	Bottle cap (septum)	4	pH ID band (blue)		0.005
AS3-606	Peristaltic pump 12 V (reversible)	1	Polycarbonate dark grey		0.35
AS3-701	Beacon mast	1	Aluminium 6061-T6, anodised	6061 tube 32 x 2.5, welded flange	0.366
AS3-702	Jal-Deep beacon base	1	Polycarbonate dark grey		0.18
AS3-703	Jal-Deep 360 deg LED lens (green normal / red alert)	1	PC lens (green = normal)		0.08
AS3-711	LTE antenna (whip, N-type)	1	Acetal (POM) black		0.12
AS3-712	Stage sensor (waterproof ultrasonic, looking up)	1	Acetal (POM) black		0.06
AS3-801	Grab handle (two-person carry)	4	Stainless steel 316 (brushed)	316 tube 22 x 1.5 bent, welded pads, into hull inserts	0.23
AS3-802	Lifting eye bolt M12 (DIN 580)	4	Stainless steel 316L		0.101

Part	Description	Qty	Material	Process	kg each
AS3-811	Cable harness (probes, stage sensor, PV, beacon)	1	Marine cable	Marine cable, IP68 glands, drip loops	0.6
AS3-812	Sampler hoses (intake + purge-to-waste)	1	Clear PVC tubing	6 mm ID food-grade tubing	0.15
AS3-901	Bolt M8 x 60 A4 + washer	16	Stainless steel 316L	A4-70	0.032
AS3-902	Bolt M10 x 70 A4 + washer	2	Stainless steel 316L	A4-70	0.06
AS3-903	Screw M6 x 25 A4	6	Stainless steel 316L	A4-70	0.01
AS3-904	Chemical anchor M12 x 130 A4	8	Stainless steel 316L	A4 threaded rod + injection mortar	0.14
AS3-905	Bolt M10 x 40 A4 + washer	4	Stainless steel 316L	A4-70	0.041

4.4 Materials and why

Material	Where	Why
316L stainless	pole, brackets, clamps, cage, spine	low carbon: welds without losing corrosion resistance; molybdenum resists pitting
6061-T6 aluminium	deck, crossbeams, carriage, mast	a third of steel's weight; weldable; anodises
LLDPE, UV-stabilised + PU foam	hulls	tough, seamless rotomoulded hulls; foam keeps it afloat if holed
UHMW-PE	rollers	self-lubricating, abrasion-resistant against silt
POM (acetal)	cartridge cap, carousel, spiders	stiff, dimensionally stable, machines cleanly
PE100 pipe D160	sensor-well sleeve	standard water pipe
EPDM; polycarbonate	buffers; IP67 pod	UV/ozone-resistant rubber; off-the-shelf sealed enclosure

4.5 Manufacturing and assembly

Rotomoulding (one mould for both hulls), laser cutting, press-brake bending and TIG welding of 316L and 6061, CNC turning and milling of rollers, collars and caps, passivation and anodising; electronics are off-the-shelf modules on a carrier PCB in an IP67 pod. Assembly follows the 12 CAD steps below (16 h for two people for the prototype, ≈ 10 h with jigs, assumption), then a 48 h soak and leak test and pH/EC calibration.

1. Pier brackets on A4 chemical anchors (on site)
2. Guide pole, stop collars, buffers (on site)
3. Twin hulls: rotomoulded, foam-filled
4. Frame: 4 crossbeams and the deck plate
5. Guide carriage and 8 UHMW rollers
6. Sensor well, guard cage, debris deflector
7. Power and control pod: battery, MPPT, controller
8. Solar canopy, 20 W
9. Autosampler: cassette, carousel, pump, valves
10. Beacon, LTE antenna, level sensor
11. Sensor cartridge: 4 probes and wiper
12. Harness, hoses, grab handles, lifting eyes

5. Engineering checks and simulation

All 14 checks are computed from the R3 CAD geometry by our own scripts (mechanical/src, results in design_checks.json). Calculated

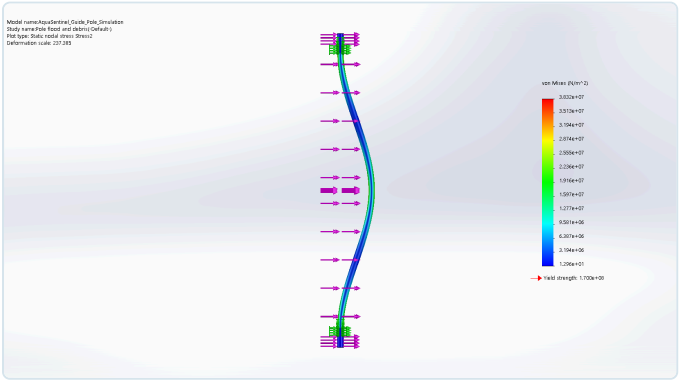
Check	Result	Criterion
Reserve buoyancy	2.27×	≥ 2.0×
Freeboard	143 mm	≥ 100 mm
Transverse GM	0.551 m	≥ 0.15 m (IMO reference)
Righting arm	positive to 60°, max 180 mm at 25°	positive to 60°
Pole safety factor, pinned	3.1 (2.66 with carriage moment)	≥ 2.0
Pole deflection, pinned	5.8 mm	≤ span/200
Jamming, rollers μ 0.05	margin 8.62	no self-locking
Jamming, seized μ 0.25	margin 1.72	no self-locking
Float lag at full load	17.7 mm	≤ 30 mm
Weathervane stability	lateral centre 351.4 mm aft	aft of pole
Pole corridor clearance	9.7 mm	≥ 5 mm
Float vs fixed parts over 2 m	no clash	none
Rub rails outermost	rub rail 12 mm outboard	outermost
Part-to-part interference	none in 101 pairs	none

5.1 Stability (GZ curve)

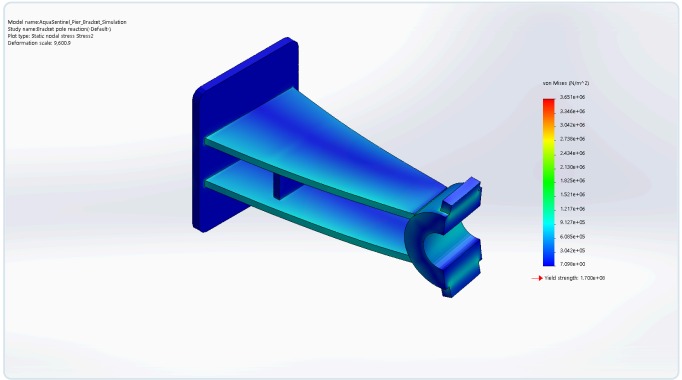
Heel °	0	5	10	15	20	25	30	35	40	45	50	55	60
GZ mm	1	46	91	137	172	180	162	140	118	96	73	50	27

5.2 Pole strength: three idealisations

Load 642 N at the float = drag $\frac{1}{2}\cdot\rho\cdot v^2\cdot Cd\cdot A = \frac{1}{2}\cdot 1000\cdot 1.5^2\cdot 1.05\cdot 0.120\ m^2$ = 142 N, plus 500 N debris applied as a quasi-static force, plus drag on the pole (81.4 N/m). Peak stress: SOLIDWORKS FEA with fixed ends 38.3 MPa (SF 4.4); pinned hand calculation 54.8 MPa (SF 3.1); pinned plus the carriage moment 64.0 MPa against 170 MPa yield → **SF 2.66, the figure we quote**. Bracket peak 3.7 MPa under the 430 N reaction. Not included: dynamic debris impact, fatigue, vortex shedding, clamp slip, welds, anchors and the bridge substrate. Calculated Simulated Not yet tested



SOLIDWORKS Simulation: pole von Mises stress



SOLIDWORKS Simulation: bracket von Mises stress

6. Electronics, power and firmware

6.1 Hardware choices

Function	Choice	Why	₹ each ex-GST
Water quality	DFRobot RS485 probes: pH SEN0708, EC SEN0707 (K = 10), optical DO SEN0680, turbidity SEN0710	one Modbus bus, IP68, 12 V; optical DO has no membrane	12,384 · 15,264 · 19,104 · 21,984
River level	JSN-SR04T ultrasonic	waterproof; separates monsoon runoff from a discharge (planned use)	265
Controller	ESP32-WROOM-32E, DS3231 RTC, microSD, MAX3485	two cores; deep sleep between readings	749 (DevKit)
Comms	SIMCom A7670C LTE Cat-1	Indian LTE bands; SMS fallback	1,900 (with carrier)
Energy	LiFePO4 12.8 V 6 Ah, 20 W panel, CN3791 MPPT	safe chemistry in heat	1,864 · 1,101 · 207
Sampler	Kamoer KPHM200 peristaltic pump (260 mL/min), two 12 V valves	sample touches only tubing; 250 mL in ≈ 58 s	1,591 · 2 × 505
Wiper; beacon	RDS3218 servo in the sealed cap; 12 V LED strobes	sweeps optical windows; readable from the bank	1,009; 2 × 762

6.2 Power budget (hardware/power_budget.csv) Estimate / assumption

Subsystem	State	Wh/day	Note
Sensor Array	Sensing Warmup Active	1.560	4x RS485 Modbus probes active during 15-min duty cycle
Edge MCU Core	Active Compute DSP	0.533	ESP32 dual-core active for sampling & anomaly detection
Cellular Modem	LTE Batch Upload	0.600	Hourly batch MQTT telemetry dispatch via SIMCom A7670C
Robotics Wiper	Motorized Optical Sweep	0.020	4 sweeps/day (every 6h) clearing optical turbidity & DO face
Robotics Autosampler	Plume Physical Capture	0.012	Triggered on anomaly (1 event/week average ~300 J)
Microcontroller Sleep	Deep Sleep RTC ULP	0.001	ESP32 in deep sleep; RTC slow timer maintains schedule
Hardware Standby Floor	Quiescent Current Floor	0.184	INA226 + buck regulator + BMS supervisory quiescent floor
Power Conversion Loss	Buck Regulator Losses	0.480	85% average conversion efficiency across buck regulators
TOTAL NOMINAL	Normal Operating Cycle	3.390	Total daily baseline consumption (3.39 Wh/day)
DESIGN TARGET MARGIN	Safety Design Target	4.070	Design baseline including 20% engineering safety margin
BATTERY CAPACITY	LiFePO4 12.8V 6000mAh	61.440	Usable energy at 80% Depth-of-Discharge (DoD)
DARK AUTONOMY DAYS	Zero Sunlight Endurance	15.1	15.1 days continuous operation without solar charging
PV MONSOON GENERATION	Overcast Monsoon Day	39.000	20W PV @ 3.0 kWh/m2/day design monsoon day (9.6x surplus)
PV ANNUAL GENERATION	Annual Average Day	68.900	20W PV @ 5.3 kWh/m2/day annual insolation (16.9x surplus)
CRISIS MODE EXTRA	Continuous 5Min Sampling	4.930	Additional consumption during prolonged chemical pollution event

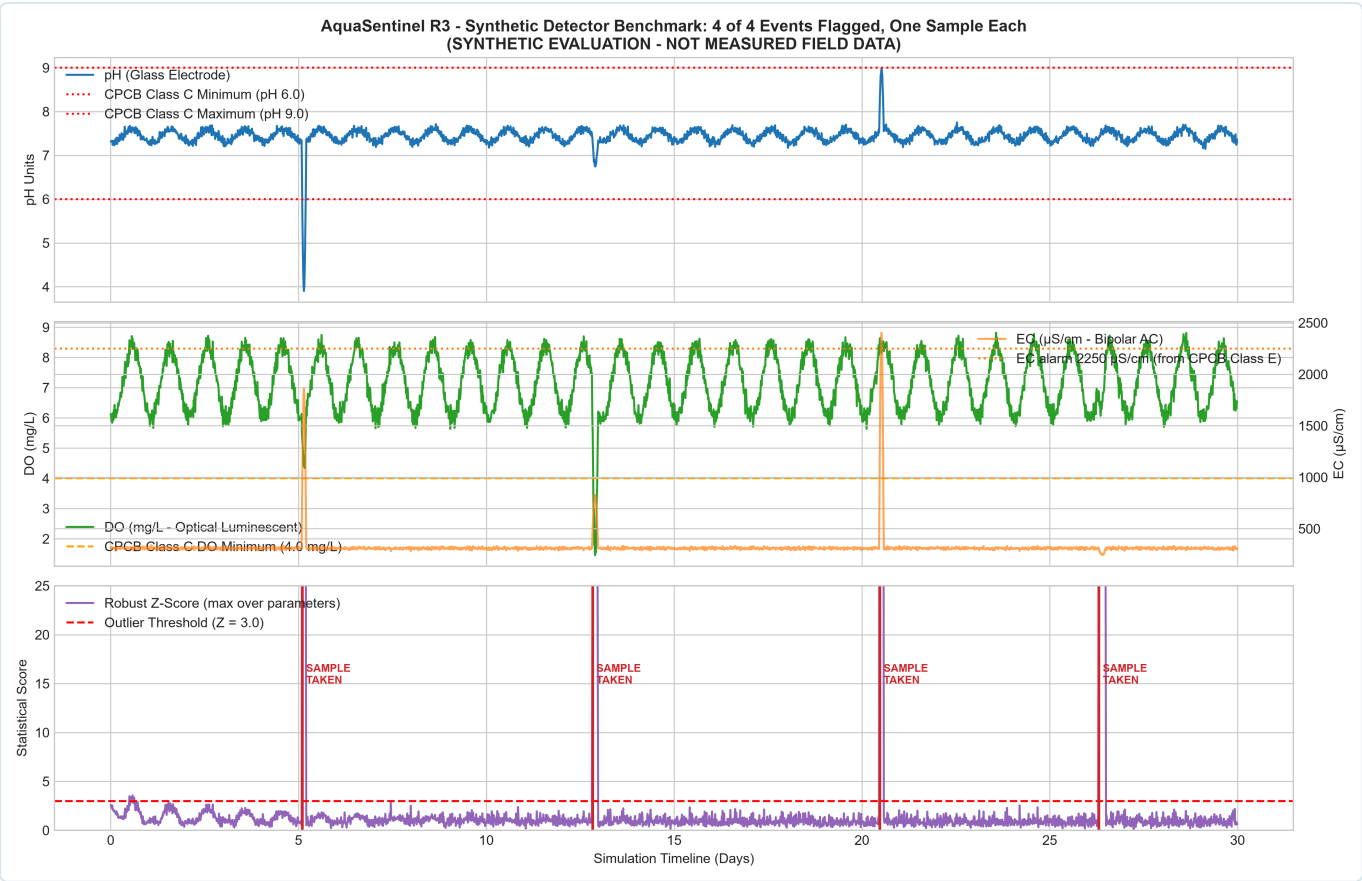
The CSV totals 3.39 Wh/day without the beacon. The deck adds an assumed beacon load (3 W strobe, 2 % duty, 12 h a night = 0.72 Wh/day): 4.11 Wh/day nominal, 4.93 with 20 % margin; 61.44 Wh usable (76.8 Wh at 80 % depth of discharge) → ≈ 12.5 days without sun (≈ 15 without the beacon). Solar: 20 W × 3.0 sun-hours on a monsoon day after ≈ 35 % losses ≈ 39 Wh/day. Pump, LTE peaks and the beacon must be measured on the bench (procedure: 06_ELECTRONICS_AND_FIRMWARE/hardware/bench_energy_test_procedure.md).

6.3 Firmware (PlatformIO, ESP32)

Modules: anomaly_engine.cpp, autosampler.cpp, dsp_filters.cpp, main.cpp, microsd_logger.cpp, sensor_interface.cpp, telemetry_comm.cpp, wiper_actuator.cpp. Unit tests: test_fail_safe and test_state_machine. Audit fixes now in the code: a failed probe read sets a sensor-fault state (never "normal water"); the learned baseline is saved to flash; one bottle per event; stale alarm state cleared when learning ends. The firmware compiles; **it has not yet run on the target hardware**. Not yet tested

7. Detection logic and analytics

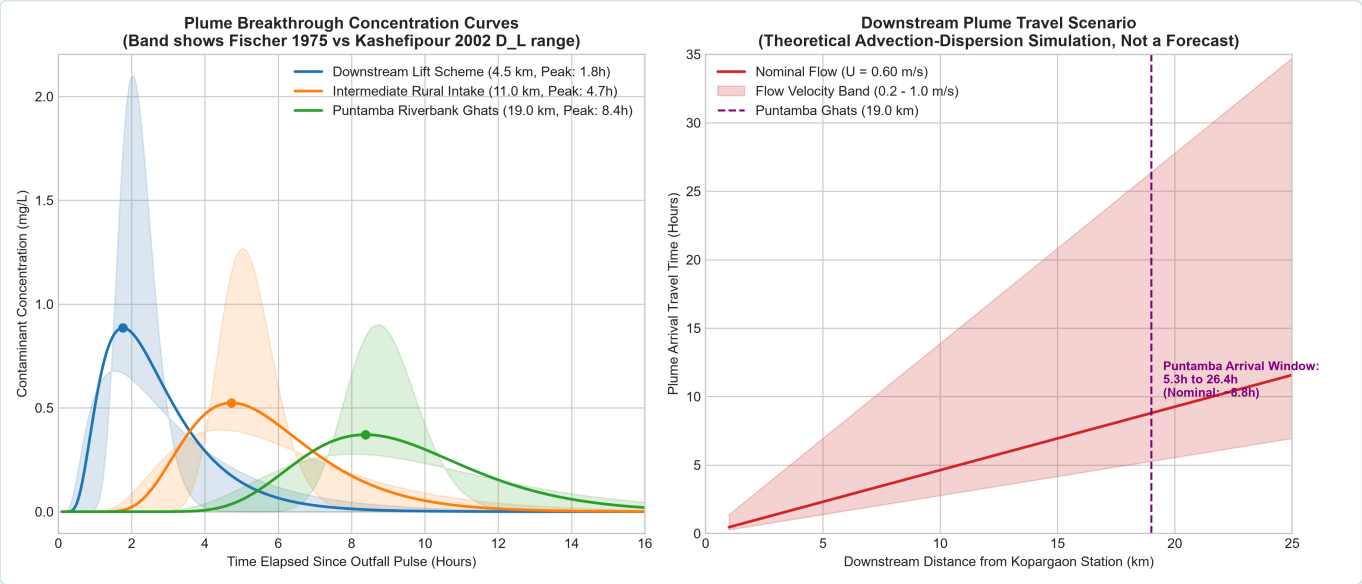
analytics/detector.py mirrors the firmware decision step; analytics/evaluate.py replays it over 30 days of 15-minute synthetic data with four injected events. Result: all four flagged within 30 minutes, one sample each, no alarms outside the events. The injected events are tens to hundreds of times the noise, so this shows the logic works, not its sensitivity or false-alarm rate in a real river. **Simulated**



Detector evaluation on synthetic data (analytics/evaluate.py)

analytics/advection_dispersion_model.py is a 1-D advection–dispersion scenario (Fischer and Kashefipour–Falconer dispersion) for a 500 kg release at 0.6 m/s on the Godavari: the peak reaches a point 4.5 km downstream about 2 hours later, which is the window an early alert could buy a water-works intake. It is a scenario, not a forecast.

Simulated



Plume breakthrough scenario (1-D model)

8. Cost: the costed bill of materials

All 66 part numbers priced from Indian listings found on 23 Oct 2026, ex-GST (input tax credit for a registered maker). Bought-in parts at listed prices; fabricated parts as CAD mass × material rate × buy factor + job-shop hours (laser ₹500/m, TIG ₹700/h, CNC ₹450/h, fitting ₹300/h) + finish. Each line is labelled Listed, Derived or Assumption. Workbook: 03_BUSINESS_COSTING_AND_BOM. **Estimate / assumption**

Subsystem	Prototype (1 unit)	At 100 units
Sensors	₹69,001	₹58,691
Electronics	₹11,484	₹8,224
Power	₹3,322	₹2,758
Actuators	₹5,603	₹5,603
Sampler	₹1,850	₹1,850
Fabricated parts	₹76,172	₹53,463
Bought mechanical	₹2,050	₹1,950
Fasteners	₹5,190	₹5,190
Assembly & test	₹6,600	₹4,800

	Prototype	At 100 units
Float	₹157,492	₹123,746
Bridge kit (one per float site)	₹29,218	₹24,309
Float + kit, ex-GST	₹186,710	₹148,055
With 18 % GST (cash)	₹220,318	—

Revised from the first estimate of the same day (₹219,348) by ₹32,638: the 10 % contingency moved once into the pilot budget, plus four design simplifications to confirm in CAD:

Part	Change	Saving, prototype	At 100
AS3-101	Pier bracket (weldment) - upper and lower identical	₹5,708	₹5,448
AS3-102	Pole clamp cap	₹3,903	₹3,855
AS3-113	Stop collar: bought 2-inch stainless clamp collar	₹2,778	₹2,169
AS3-301	Guide carriage - fixed half (weldment)	₹268	₹268
AS3-302	Guide carriage - opening gate (weldment)	₹183	₹183

The old team sheet (v2 layout) totalled ₹1,38,000 (FDC deck: ₹1.34 lakh): its probes were priced right, but the heavier R3 structure, fasteners, labour and contingency were missing or too low.

8.1 Every priced line

Part	Item	Qty	₹ each	₹ total	Basis	Source id
AS3-420	pH probe, DFRobot SEN0708 (RS485, IP68)	1	₹12,384	₹12,384	Listed	S1, S4, S5
AS3-420	EC probe K=10, DFRobot SEN0707 (RS485, IP68)	1	₹15,264	₹15,264	Listed	S2, S4, S5
AS3-420	Optical DO probe, DFRobot SEN0680 (RS485, IP68)	1	₹19,104	₹19,104	Listed	S1, S4, S5
AS3-420	Turbidity probe 0-1000 NTU, DFRobot SEN0710 (RS485)	1	₹21,984	₹21,984	Listed	S3, S4, S5
AS3-712	River-level sensor JSN-SR04T (waterproof ultrasonic)	1	₹265	₹265	Listed	S7
AS3-515	ESP32-WROOM-32E (DevKitC for the prototype)	1	₹749	₹749	Listed	S8
AS3-515	LTE Cat-1 modem SIMCom A7670C (on carrier board)	1	₹1,900	₹1,900	Listed	S9
AS3-515	RS485 transceiver MAX3485	1	₹325	₹325	Listed	S19
AS3-515	Real-time clock DS3231 module	1	₹105	₹105	Listed	S19
AS3-515	Power monitor INA226 module	1	₹285	₹285	Listed	S19
AS3-515	PWM driver PCA9685 module (beacon, wiper, valve)	1	₹300	₹300	Assumption	—
AS3-515	microSD socket + 16 GB industrial card	1	₹510	₹510	Assumption	—
AS3-515	Buck converters 3.3 V / 4.0 V / 5 V	3	₹150	₹450	Assumption	—
AS3-515	MOSFET load switches, TVS, fuse, connectors, passives	1	₹600	₹600	Assumption	—
AS3-515	Carrier PCB, 2-layer (5 boards incl. shipping)	1	₹2,100	₹2,100	Derived	S21
AS3-511	IP67 polycarbonate enclosure (pod body + lid)	1	₹1,500	₹1,500	Listed	S18
AS3-811	IP68 cable glands M16	10	₹70	₹700	Derived	S22
AS3-711	4G LTE antenna + IP67 bulkhead pigtail	1	₹560	₹560	Derived	S17
AS3-811	Marine cable harness (15 m) + connectors	1	₹1,400	₹1,400	Assumption	—
AS3-514	LiFePO4 12.8 V 6 Ah with BMS	1	₹1,864	₹1,864	Listed	S10
AS3-521	Solar panel 20 W, 12 V	1	₹1,101	₹1,101	Listed	S11
AS3-516	MPPT solar charger CN3791 (set to 14.4 V for LiFePO4)	1	₹207	₹207	Listed	S12
AS3-516	Battery fuse, holder and isolator	1	₹150	₹150	Assumption	—
AS3-606	Peristaltic pump 12 V, 260 mL/min (Kamoer KPHM200)	1	₹1,591	₹1,591	Listed	S13
AS3-606	Solenoid valves 12 V (sample / purge-to-waste)	2	₹505	₹1,010	Listed	S14
AS3-603	Carousel drive: N20 12 V gear motor	1	₹189	₹189	Listed	S20
AS3-603	Carousel index sensor (optical slot) + bearing	1	₹280	₹280	Assumption	—
AS3-411	Wiper servo, metal gear, waterproof (RDS3218)	1	₹1,009	₹1,009	Listed	S15
AS3-703	Beacon: 12 V waterproof LED strobes, green + red	2	₹762	₹1,524	Listed	S16
AS3-604	Sample bottles, 250 mL HDPE wide mouth	4	₹72	₹290	Listed	S23
AS3-605	Septum caps	4	₹30	₹120	Assumption	—
AS3-812	Food-grade tubing (intake, purge)	3	₹80	₹240	Assumption	—
AS3-601	Insulated sampler cassette + lid (bought cool box, modified)	1	₹1,200	₹1,200	Assumption	—
AS3-101	Pier bracket (weldment) - upper and lower identical	2	₹6,404	₹12,808	Derived	S25, S29-S32
AS3-102	Pole clamp cap	2	₹1,639	₹3,279	Derived	S25, S29-S32
AS3-103	Stage target plate	1	₹680	₹680	Derived	S25, S29-S32
AS3-111	Guide pole 2in Sch 40S SS316L	1	₹7,690	₹7,690	Derived	S24, S29-S32
AS3-211	Crossbeam SHS 40x40x3	4	₹797	₹3,186	Derived	S26, S29-S32
AS3-213	Equipment deck plate	1	₹4,413	₹4,413	Derived	S26, S29-S32
AS3-301	Guide carriage - fixed half (weldment)	1	₹7,483	₹7,483	Derived	S26, S29-S32
AS3-302	Guide carriage - opening gate (weldment)	1	₹4,971	₹4,971	Derived	S26, S29-S32
AS3-311	Concave guide roller UHMW-PE	8	₹279	₹2,233	Derived	S33, S29-S32
AS3-313	Gate hinge pin D10	1	₹390	₹390	Derived	S24, S29-S32
AS3-314	T-handle latch pin (tool-free)	2	₹397	₹794	Derived	S24, S29-S32
AS3-321	Carriage brace struts (pair)	1	₹780	₹780	Derived	S24, S29-S32
AS3-402	Hatch flange ring (quarter-turn seat)	1	₹1,177	₹1,177	Derived	S25, S29-S32
AS3-403	Sensor guard cage	1	₹3,428	₹3,428	Derived	S24, S29-S32
AS3-404	V debris deflector	1	₹404	₹404	Derived	S33, S29-S32
AS3-411	Cartridge cap + wiper drive housing	1	₹2,036	₹2,036	Derived	S33, S29-S32
AS3-412	Cartridge lift handle	1	₹235	₹235	Derived	S24, S29-S32
AS3-413	Cartridge spine + wiper shaft	1	₹1,273	₹1,273	Derived	S24, S29-S32

Part	Item	Qty	₹ each	₹ total	Basis	Source id
AS3-414	Probe holder spiders (x2)	1	₹578	₹578	Derived	S33, S29-S32
AS3-415	Optical wiper arm + brush pads	1	₹631	₹631	Derived	S24, S29-S32
AS3-417	Sampler intake strainer + riser	1	₹448	₹448	Derived	S24, S29-S32
AS3-501	Pod mounting rails (pair)	1	₹533	₹533	Derived	S26, S29-S32
AS3-513	Pod mounting plate	1	₹1,139	₹1,139	Derived	S26, S29-S32
AS3-523	Canopy stanchions + hinge / pin blocks	1	₹1,737	₹1,737	Derived	S26, S29-S32
AS3-603	Bottle carousel (lift-out cassette)	1	₹1,118	₹1,118	Derived	S33, S29-S32
AS3-701	Beacon mast	1	₹690	₹690	Derived	S26, S29-S32
AS3-801	Grab handle (two-person carry)	4	₹480	₹1,920	Derived	S24, S29-S32
AS3-201	Pontoon hull, foam filled (prototype: welded PE sheet; production: rotomoulded LLDPE)	2	₹4,934	₹9,868	Derived	S27, S33, S34
AS3-113	Stop collar: bought 2-inch stainless clamp collar	2	₹500	₹1,000	Assumption	—
AS3-401	Sensor-well sleeve, PE100 D160 pipe (1 m bought)	1	₹249	₹249	Listed	S35
AS3-112	Pole top cap (POM)	1	₹150	₹150	Assumption	—
AS3-114	Stop buffer ring, EPDM	2	₹150	₹300	Assumption	—
AS3-202	Bow bumper pad, EPDM	2	₹150	₹300	Assumption	—
AS3-203	Hull vent / test plug	2	₹50	₹100	Assumption	—
AS3-212	Beam end cap	8	₹15	₹120	Assumption	—
AS3-421	Probe ID bands (4 colours)	4	₹20	₹80	Assumption	—
AS3-901	Bolt M8 x 60 A4 + washer	16	₹30	₹480	Assumption	—
AS3-902	Bolt M10 x 70 A4 + washer	2	₹45	₹90	Assumption	—
AS3-903	Screw M6 x 25 A4	6	₹10	₹60	Assumption	—
AS3-905	Bolt M10 x 40 A4 + washer	4	₹35	₹140	Assumption	—
AS3-312	Roller axle M10 shoulder bolt + nyloc	8	₹120	₹960	Assumption	—
AS3-802	Lifting eye bolt M12 DIN 580	4	₹250	₹1,000	Assumption	—
AS3-904	Chemical anchor stud M12 x 130 A4	8	₹120	₹960	Assumption	—
AS3-904	Injection mortar cartridge	1	₹1,500	₹1,500	Assumption	—
—	Assembly and wiring (2 people)	16	₹300	₹4,800	Derived	S30
—	Leak test, 48 h soak, calibration (pH buffers, EC 1413 uS/cm)	1	₹1,800	₹1,800	Assumption	—

9. Business model, market and the ask

9.1 Monitoring as a service (proposal) Estimate / assumption

Per float, 100-unit cost	₹
Up front: float + kit + installation	₹163,055
Running cost a year: monthly visit (12 × ₹1,500), 4 lab samples (₹3,500 each), consumables, LTE data, cloud	₹44,600
Fee	₹12,000 a month = ₹144,000 a year, 5-year contract
Payback of hardware + installation	≈ 20 months
5-year revenue / costs / margin	₹7.2 lakh / ≈ ₹3.86 lakh / ≈ 46 %
Same on a 3-year contract	≈ 31 % margin (so we ask for 5 years or a setup fee)
Cost to serve a year incl. hardware spread over 5 years	₹77,211
Outright purchase option	≈ ₹300,000 installed + ₹36,000 a year maintenance

9.2 Customers

Pollution control boards (MPCB): catch discharges, know when to sample. Water-works: close an intake before a plume. Irrigation / WRD: continuous record with river level. Industrial estates (MIDC): an independent downstream record. No letter of interest yet: a gap to close in the pilot.

9.3 Market (illustrative)

Scope	Floats (3 per counted site)	Revenue a year at ₹1.44 lakh
Maharashtra, 54 sites	162	≈ ₹2.33 crore
India, 296 sites	888	≈ ₹12.8 crore
All 4,922 CPCB monitoring locations (one each)	4,922	≈ ₹70.9 crore

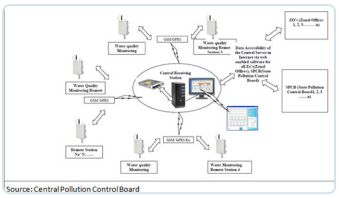
Context: India water analytical instruments $150.4 M (2024) → $253.4 M by 2033, 6 % a year (IMARC). Three floats per site is an assumption.

9.4 The ask: a 3-float, 6-month Godavari pilot

Item	₹
3 floats + bridge kits at prototype cost	₹560,130
Lab validation: 60 samples at Rs 3,500 (S36)	₹210,000
Installation at 3 sites	₹45,000
6 months of field service and data	₹32,400
Contingency 10 %	₹84,753
Total	₹932,284

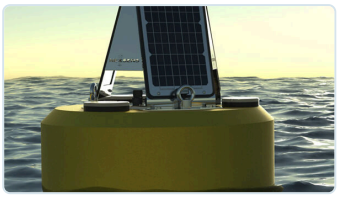
Funding routes: NIDHI-PRAYAS (up to ₹10 lakh), CSR, incubator pre-seed. The pilot measures detection against the lab, the false-alarm rate, the real cost to serve and willingness to pay.

10. Prior art and competition



CPCB real-time network (CSE)


YSI EXO2 sonde (YSI)


NexSens CB-450 buoy (NexSens)



Teledyne ISCO 6712 sampler (RS Hydro)

These notes back deck v4's slides 12–13. "n.p.v." means not publicly verified. This is a competitive-landscape review, not a novelty, patentability or freedom-to-operate opinion.

1. Existing solutions reviewed

Solution	Type	Source
CPCB real-time water quality stations, river Ganga	deployed system, India	CPCB overview, CSE
YSI EXO2 multiparameter sonde (Xylem)	commercial product	YSI EXO2
NexSens CB-450 data buoy	commercial product	NexSens CB-450
Teledyne ISCO 6712 portable sampler	commercial product	RS Hydro / Teledyne ISCO
CN117446093B, US8352213	patent documents (not proof of products)	CN117446093B, US8352213

2. Verified strengths and trade-offs

CPCB stations

- **Strengths:**
  - about 17 parameters, including BOD, COD, ammonia, TOC and level;
  - readings every 15 min, sent over GSM/GPRS to a central server;
  - 36 stations commissioned in 2017, with 40 more planned (CPCB).
- **Trade-offs:**
  - instruments cost about ₹35–40 lakh per station (CSE), so stations are few;
  - station layout and any physical sampling: n.p.v.

YSI EXO2

- **Strengths:**
  - 7 ports: up to 6 sensors plus a central wiper;
  - RS-485, SDI-12 and Modbus output;
  - about 90 days on 4 D-cells at 15-minute logging (YSI; depends on configuration).
- **Trade-offs:**
  - it's an instrument, so it needs a mount, a logger and telemetry for live alerts;
  - price on request.

NexSens CB-450

- **Strengths:**
  - carries sondes and sensors;
  - 3 × 15 W solar panels;
  - LTE or Iridium through the X3 logger;

    - 86 cm hull, 56 kg.
- **Trade-offs:**
    - moored with anchors (3 eye-nuts);
    - a beacon or sampler isn't part of the listed product;
    - price not listed.

ISCO 6712

- **Strengths:**
    - from 24 × 1 L bottles to a single 5.5 gal container;
    - time, flow or event sampling;
    - triggers on up to 16 inputs, including SDI-12 sondes;
    - optional modem dial-out alarms.
- **Trade-offs:**
    - a bank or manhole unit with an intake hose;
    - takes samples but doesn't log continuous readings itself.

Patents

- **CN117446093B:** floating detection units held within an area by fixed rods and ropes.
- **US8352213:** a pier-mounted rod that slides vertically, for riverbed elevation.
- No exact match was found for a float on a pier-clamped guide pole carrying water-quality sensors. The search was English-only.

3. AquaSentinel's defensible differentiators (narrow; all designed, not validated)

The differences are architectural. AquaSentinel combines the following in one unit, at a prototype estimate of ₹1.87 lakh including the bridge kit:

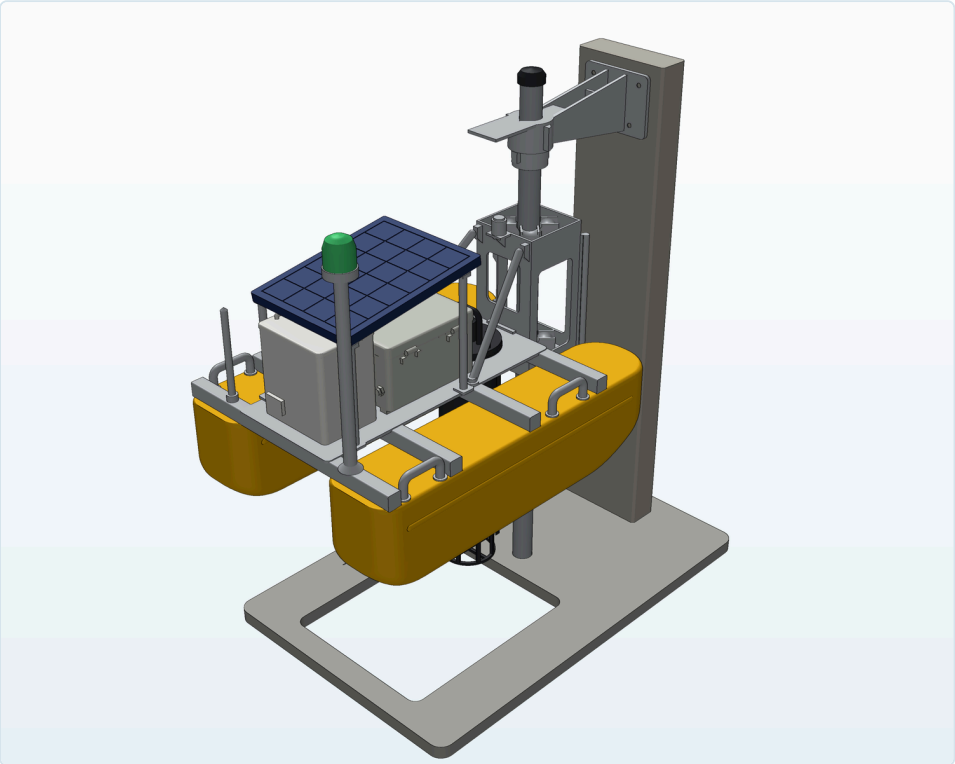
- **Guide pole:** a float on a guide pole clamped to a bridge pier, which follows river level without a mooring (2 m travel, calculated).
- **On-board decision:** fixed limits plus a learned baseline (tested on synthetic data only).
- **Alerts:** a beacon visible from the bank, plus LTE/SMS alerts.
- **Sampling:** one 250 mL bottle per event, 4 bottles in all.

4. Similarities, evidence gaps, claims to verify

- **Every capability exists separately.** Real-time 15-minute monitoring (CPCB), sondes with wipers (YSI), solar buoys with LTE (NexSens) and sonde-triggered samplers (ISCO). A buoy, a sonde and a triggered sampler could be integrated by others.
- **Not checked:**
    - Chinese-language patent search;
    - other buoy and pontoon makers (for example Xylem/YSI EMM buoys, Libelium, Eruvaka), and whether any already sells a pier-guided float;
    - competitor prices;
    - CPCB station layout and sampling.
- **Ours, still to prove:**
    - level tracking and roller friction in a tank;
    - detection on real data;
    - power with the beacon;
    - the sampler's sample integrity;
    - the cost through quotes.

11. Status, roadmap and risks

Level	What it shows	Status
0 · Digital design	CAD, calculations and FEA; logic flags 4 of 4 simulated spills	Done
1 · 1:5 scale model	Layout and mechanism; float slides 65 mm up its pole (325 mm full scale, not the full 2 m); cartridge lifts out. No electronics inside	Print files ready
2 · Electronics bench demo	ESP32 + a real probe: read, decide, act (light, pump, alert)	Next, before 19 Nov
3 · Tank test	Full-size float on a 2-inch pole: draft, 2 m travel, roller friction	After the finale
4 · Godavari pilot	30 days at Kopargaon; alerts checked against lab samples	After permits



Render of the 1:5 model (not a photo)

Dates

India Semi-Final 24–25 Oct 2026 (online, 8-minute pitch). India Geo Finale 19–20 Nov (in person). International Finale 14–17 Feb 2027, 3DEXPERIENCE World, Nashville.

Risk	Why it matters	How we handle it
Flood above 2 m of travel	float held under at the top stop	gauge data (CWC/WRD); longer travel or a breakaway link; lift out on flood warnings
Bridge-pier wake	unsteady yaw and cyclic loads	observe in tank and on site; side-face mounting if needed
Probe fouling and drift	false or missed alarms	wiper; monthly service with before/after readings; lab comparison
False alarms	users stop trusting alerts	2-of-3 rule, 7-day learning, river level to tell runoff; pilot measures the rate
Level sensor blind zone	no level near high water	compare a pressure transducer and a downward radar
Sample is not legal evidence	Water Act §21 needs the Board's procedure	a triage sample that triggers the Board's formal sampling
Beacon colours	red/green read as navigation marks	ask the waterway authority; yellow or shielded light

Risk	Why it matters	How we handle it
Permits and bridge approval	no pilot without them	bridge owner, river authority and MPCB; structural engineer for the attachment
Theft or damage	loss of a float	under-bridge placement, sealed pod, tamper-seal hasp on the sampler
Probe supply	44 % of the float cost from one maker	any RS485 Modbus probe can replace it

12. Engineering opportunities (proposals, not claims)

Each item below is a proposal that would strengthen AquaSentinel; none is built or tested. **Estimate / assumption**

Opportunity	Why it helps	Effort
Use river level in the decision (stage-aware baselines)	separates monsoon runoff (EC, turbidity change with level) from a discharge; already planned	firmware + pilot data
Data-quality flags on every reading	stuck values, out-of-range, wiper failure and drift flagged instead of alarmed; strengthens 'never normal on a fault'	firmware
Alert queue with retry and SMS fallback, logged	alerts survive network outages; proves notification latency separately from detection latency	firmware + bench test
Measured energy budget	replaces the 12-day estimate with a measured one (procedure already written)	bench test, 1 day
Event chain-of-custody log	bottle number, timestamps, temperature and retrieval record for each sample; supports the Board's formal sampling	firmware + form
Pressure transducer as a second level source	removes the ultrasonic blind zone near high water	≈ ₹2–5k part (estimate), CAD change
Breakaway link or lift-out procedure	protects the float and the bridge when a flood exceeds 2 m of travel	CAD + procedure
Dynamic debris-impact and anchor check	closes the biggest structural gap behind the 2.66 figure	hand calc / FEA + structural engineer
Optional oil-sheen or fluorescence sensor	covers oil spills that EC, pH, DO and turbidity miss, where a site needs it	sensor choice + cost study
Lite float (pH, EC, temperature, level)	≈ ₹113,483 prototype, ₹86,274 at 100: an entry product for industrial outfalls	configuration only

13. The 8-minute pitch script

about 140 words a minute; about 880 spoken words plus the 60 s video = about 7:30, leaving 30 s for pauses and clicks. Slide numbers are the deck's; the cover is slide 1. Say "designed", not "built"; "a sample for the lab", never "evidence".

Slide	Time	Speaker	Topic
1	before the timer	Soham	Cover
2	0:00-0:40	S1 Soham	The problem
3	0:40-1:45	S1, then the video	Marketing video
4	1:45-2:25	S2 Tanmay or Bhakti	What it is
5	2:25-3:10	S2	How it works
6	3:10-3:50	S3 (whoever knows the CAD best)	Design checks and simulation
7	3:50-4:20	S3	Materials and manufacturing
8	4:20-5:30	S1 Soham	Customers, cost and business model
9	5:30-6:15	S2	Where we are, honestly
10	6:15-6:45	S1 Soham	Prior art
11	6:45-7:15	S1 Soham	How AquaSentinel differs
12	7:15-7:50	S1 Soham	Close

Slide 1 · Cover (before the timer, Soham)

Show the cover while we are introduced. When the 8-minute timer starts, click once to slide 2. Animations play by themselves. In Q&A, type 13 and press Enter for the questions menu (slides 10-11 are prior art; 14-22 are backups A1-A9).

Slide 2 · The problem (0:00-0:40, S1 Soham)

> It's two in the morning on the Godavari. A tanker opens its valve, and by sunrise the river looks normal again. Nobody saw it, and the next official sample may be weeks away. India's latest CPCB assessment lists 296 polluted river stretches and locations, and Maharashtra, our state, has the most: 54. Most monitoring points are sampled by hand, monthly or quarterly. Real-time stations exist, but they cost tens of lakhs each, so there are very few. A spill that lasts a few hours falls through the gap. We are Team Sanjivani Hero's Journey, and AquaSentinel is our answer.

Slide 3 · Marketing video (0:40-1:45, S1, then the video)

Say: 'Here is AquaSentinel in sixty seconds.' Click the video to play it. If it does not start, play AquaSentinel_Marketing_60s.mp4 from 02_PRESENTATION_SEMIFINAL and keep Zoom's Share sound on.

Slide 4 · What it is (1:45-2:25, S2 Tanmay or Bhakti)

> AquaSentinel is a river robot designed to sense, think and act on its own. It senses pH, conductivity, dissolved oxygen, turbidity, temperature and river level. An ESP32 on board decides every fifteen minutes, with no cloud needed. When the water changes, the beacon turns red, an alert goes to the water-works, and a pump fills a sample bottle for the lab. It is an early-warning system, not a chemical identifier. The table shows each objective, and honestly, how far it is proven.

Slide 5 · How it works (2:25-3:10, S2)

> How does it stay in place without a motor? A two-inch stainless pole is clamped to a bridge pier, so nobody works in the riverbed. Eight rollers in two tiers let the river itself carry the float up and down

> through two metres, and its side area sits behind the pole, so the current turns it like a weathervane. Every fifteen minutes it checks hard limits, like pH outside six to nine or oxygen below four, and a baseline it learns for every hour of the day. Two unusual readings out of three make an event.

Slide 6 · Design checks and simulation (3:10-3:50, S3 (whoever knows the CAD best))

> We checked every requirement on the CAD model itself. The float weighs 48.5 kilograms by our estimate, floats with 143 millimetres of freeboard and more than twice the buoyancy it needs, and rights itself up to sixty degrees. For a flood we designed for 1.5 metres per second plus a 500-newton debris load. Simulation with fixed ends gives a safety factor of 4.4; we quote our most conservative case, 2.66. That is a static case: impact and anchors are still to verify.

Slide 7 · Materials and manufacturing (3:50-4:20, S3)

> R3 is a 66-part SOLIDWORKS assembly with a real material on every part: two polyethylene hulls from one rotomould, aluminium deck and carriage, a stainless pole, and off-the-shelf electronics, so a local fabricator can build it. CAD also caught our two biggest mistakes: the pole once ran through the electronics pod, and the real mass came out at 48.5 kilograms, not 25.

Slide 8 · Customers, cost and business model (4:20-5:30, S1 Soham)

> Who pays? Pollution control boards, water-works protecting their intakes, irrigation departments and industrial estates. We priced all 66 parts from Indian suppliers: one float costs about 1.6 lakh rupees before GST, plus a 29-thousand-rupee bridge kit; at a hundred units, about 1.5 lakh together. But a board doesn't buy it. It pays twelve thousand rupees a month on a five-year contract, like station maintenance tenders. That pays back the hardware in about twenty months, with a margin of about 46 percent. One real-time station's instruments alone cost thirty-five to forty lakh. We start with three floats at Kopargaon, then Maharashtra's 54 polluted sites.

Slide 9 · Where we are, honestly (5:30-6:15, S2)

> Where are we, honestly? AquaSentinel is designed, calculated and simulated; it has not been in the water yet. On thirty days of synthetic data, our logic flagged all four injected spills within thirty minutes. That proves the logic, not field accuracy. Power, including the beacon, is about twelve days without sun, an estimate we will measure. Before the finale: the 1:5 model in water and the logic on an ESP32 with a real probe. Then a full-size tank test, and after permits, a thirty-day pilot at Kopargaon checked against lab samples.

Slide 10 · Prior art (6:15-6:45, S1 Soham)

> We are not the first to watch rivers. CPCB's real-time stations read seventeen parameters every fifteen minutes. YSI's sonde is the reference instrument, NexSens floats sondes on solar buoys, and Teledyne ISCO's samplers can be triggered by a sonde. Each does its job well. But each does one part of the job.

Slide 11 · How AquaSentinel differs (6:45-7:15, S1 Soham)

> Our difference is how we combine them: a float on a pier-mounted guide pole that follows the river without a mooring, a decision made on board, a light people on the bank can see, and one sample bottle per event, in one unit under two lakh. Each capability exists separately; the combination is what we must now prove in water.

Slide 12 · Close (7:15-7:50, S1 Soham)

> A spill lasts hours. A sample comes once a month. AquaSentinel is designed to close that gap: ninety-six readings a day, a decision on the spot, a red light on the bank, and a sample that tells the lab where to look. We are Soham, Shriraj, Bhakti and Tanmay, guided by Dr. Kiran Wakchaure. Thank you. We'd love your questions.

14. Jury questions and answers

Version 1, 19 Oct 2026. Every answer is built from the repository and the audit of 18 Oct; the evidence column says where to look if a juror pushes. When an answer admits a limit, it gives the next step. Never trade honesty for a smoother answer (rules, Art. 10).

How to answer

1. **Answer in 20 seconds:** the number or fact first, then the reason. Five minutes is about 8–10 questions.
2. **If it's a limit, say it plainly** and give the next step ("Not yet. We'll measure it in the tank test.").
3. **Never guess a number.** Say "I'll confirm that" and write it down.
4. **Hand over by area:** S1 for business and story, S2 for software, sensing and power, S3 for mechanical, CAD and FEA.
5. **Words to avoid:**
    - "evidence", "forensic" or "legal proof" about the sample (say "a triage sample for the lab")
    - "deployment-ready"
    - "4.4 safety factor" without its context
    - "25 times cheaper" without the caveat
    - "tested" or "works" for anything not yet tested
    - "Class C" for the EC limit
    - "AI" for the rule-based engine

A. Theme and concept

#	Question	Say	Evidence
A1	Why is this Robotics and not just an IoT sensor?	It closes the loop on its own. It senses six values, decides on board every 15 minutes, and acts physically through a pump, a pinch valve, a four-bottle carousel, a wiper and a beacon. And it keeps station through 2 m of river rise with a passive guided mechanism, so it needs no drive motor (small actuators only clean and sample).	deck slides 5–8; cost sheet "Robotics Autosampler", "Robotics Wiper"
A2	What's new about it?	Existing options don't combine three things: continuous readings, low cost, and a physical sample on alert. AquaSentinel does, on a float held by a bridge pier, so there's no mooring and it's serviced from the deck.	deck slide 2; TECHNICAL_RESEARCH.md §5
A3	Why a float on a pole instead of a buoy, or a pipe on the pier?	A float keeps the probes at a constant depth as the river rises 2 m, and the pole keeps it on station and lets us lift it out. A fixed pipe leaves the probes dry at low water; a moored buoy brings mooring and theft problems.	research §5

B. Status and honesty

#	Question	Say	Evidence
B1	Have you built or tested it?	Not yet. It's fully designed: a 66-part CAD model, hand calculations and SOLIDWORKS Simulation. The decision logic has been replayed on 30 days of synthetic data. Before the finale we'll float our 1:5 scale model and run the logic on the ESP32 with a real probe on the bench. After that comes a full-size tank test of draft, 2 m travel and roller friction, then a 30-day pilot at Kopargaon with permission.	FDC post; design_checks.json

#	Question	Say	Evidence
B1a	(Use only once the 1:5 model is printed.) Have you built anything physical?	We've 3D-printed a 1:5 scale model straight from the CAD to check the arrangement and show the mechanism: the float slides up its pole, the sensor cartridge lifts out, and the beacon goes green to red. It isn't a working prototype and has no electronics. Before the finale we'll float it in a tank and show the electronics bench demo; the full-size tank test comes after the finale.	scale model guide
B2	Does the firmware run on the hardware?	Not yet on hardware. The firmware now compiles with our audit's fixes: a dead probe raises a sensor fault, each event takes one bottle, and the learned baseline survives a power cut. Its 26 unit tests are ready to run on the board; the bench demo before the finale is the first hardware test.	CODE_REVIEW.md, firmware README (branch `bench-demo`)
B3	Your FDC deck said a safety factor of 4.4. Now 2.66?	4.4 is FEA with both pole ends fixed. Our conservative hand calculation treats the ends as pinned and adds the carriage moment: 64 MPa against a 170 MPa yield, so 2.66, still above our 2.0 target. We quote the conservative number.	pole_carriage_couple.py
B4	How good is "4 of 4 spills within 30 minutes"?	It's honest but easy. The simulated spills were large compared with the noise. Moderate events and the real false-alarm rate can only come from pilot data with lab samples alongside, and that's what the pilot is designed to measure.	ANALYTICS_AND_DATA_REVIEW.md §1.1, §2.4

C. Mechanical

#	Question	Say	Evidence
C1	What if the river rises more than 2 m?	Then the float reaches the top stop and would be held under. That's our biggest unknown, because we don't yet have Kopargaon's flood-stage record. Before the pilot we'll get the CWC or state gauge data and either lengthen the travel, add a breakaway link, or lift the float out on flood warnings.	audit "ten things to know" #5
C2	What about debris and water hyacinth?	We designed for a 500 N debris snag on top of 142 N of drag at 1.5 m/s, so 642 N at the float. There's a debris deflector and a perforated stainless cage around the probes; weed on the probes is handled by the wiper and service visits.	design_checks.json
C3	Can the carriage jam on the pole?	We used the drawer rule: it can't self-lock while the load's offset is less than the roller spacing divided by twice the friction coefficient. With 350 mm between roller tiers the margin is 8.6, and still 1.7 if the rollers seize and slide.	design_checks.json "carriage"
C4	Can it capsize?	Transverse GM is 0.55 m. The righting arm peaks at 180 mm at 25° and is positive at every angle we computed, up to 60°. Reserve buoyancy is 2.27 times its weight, with 143 mm of freeboard.	design_checks.json "hydrostatics"
C5	Where do 1.5 m/s and 500 N come from?	They're design assumptions for a monsoon flood with debris, not site measurements. Site data and the pilot will confirm them; at these loads the structure still has margin.	design_checks.json "hydrodynamics"
C6	Won't the bridge pier's wake shake it?	It sits in the pier's near wake, so we expect unsteady yaw. The tank test or a site observation will show how much. If needed, we mount it on the side face of the pier, clear of the wake.	research §3.4
C7	How do you install and remove it?	Two brackets bolted to the pier, one 2-inch stainless pole, slide the float on and close the gate (a hinge and two T-pins, no tools). It weighs 48.5 kg, so it's a two-person lift with four handles and four lifting eyes.	deck slides 4, 8, 12

D. Sensing and decision

#	Question	Say	Evidence
D1	How accurate are the probes?	They're industrial RS485 probes: pH ±0.15, EC ±1 % of full scale, DO ±3 %, turbidity ±5 %. At river levels that's coarse, so we detect changes against a learned baseline and calibrate against lab samples in the pilot.	research §2.1
D2	How do you avoid false alarms?	Hard limits act immediately. Everything else needs 2 of the last 3 readings outside a baseline learned for each hour of the day, using a robust z-score. The baseline learns for the first 7 days.	ANALYTICS §2.1
D3	How do you tell monsoon silt from pollution?	Today the engine would flag a big turbidity surge either way. Our next step is to use the river-level record: turbidity rising with the river suggests runoff, while EC or pH changing without a rise suggests a discharge. The pilot will tell us how well that separates them.	ANALYTICS §1.1
D4	Why not AI or ML?	With no field data yet, a statistical baseline is explainable and runs on the ESP32. Once the pilot gives labelled data we'll benchmark against EPA's open CANARY event-detection tool and against learned models.	research §3.2
D5	What if a probe fails or drifts?	A failed reading must never look like clean water. Our own audit found that gap, and the fix is now written in our firmware, ready for the bench test: a failed read sets a sensor-fault state, never "normal". Drift is checked at monthly service visits with before-and-after-cleaning readings, following USGS practice.	audit "ten things" #1; research §3.1
D6	How do you measure the river level?	An ultrasonic sensor looks up at a plate on the top bracket. Our analysis showed it would sit inside its blind zone near high water, so before the pilot we're comparing a pressure transducer and a downward radar on the bracket.	research §2.4
D7	What about fouling?	A wiper sweeps the optical DO and turbidity windows every 6 hours, and the whole probe cartridge pulls up from the deck for cleaning. The DO sensing film lasts about a year; turbidity is recalibrated about every 90 days.	research §2.1; power budget
D8	What if the power cuts out and it reboots?	It keeps what it learned. Our audit found the baseline lived only in memory that a power cut wipes, so the firmware now saves the learned baseline, the learning progress and the alarm state to flash, and checks why it restarted at boot. After a power cut it carries on instead of learning for another 7 days. This is written and compiles with unit tests; the bench test is its first run on hardware.	CODE_REVIEW.md F5; firmware `anomaly_engine.cpp` (commit `64e6289`)

E. Sampler and evidence

#	Question	Say	Evidence
E1	Is your sample legal evidence?	No, and we don't claim it. Under Section 21 of the Water Act a legal sample needs the Board's own procedure: notice, splitting and sealing in the occupier's presence. Our bottle is a triage sample. It tells the Board when and where to take that formal sample, and gives the lab a first look.	research §1.2
E2	How long can the sample wait?	Not long. BOD samples should be cooled to 6 °C or below and analysed within 48 hours, so an alert must trigger collection within hours. We'll agree bottles and preservation for each analyte with a lab partner.	research §1.3
E3	Won't one sample contaminate the next?	Each fill starts with a 20-second purge to waste, then fills a separate 250 mL bottle on a four-position carousel. Carry-over isn't measured yet; that's a bench test.	cost sheet G-01 to G-04
E4	If a spill lasts hours, do you use up all four bottles?	No: one bottle per event. Our own check found the first version re-triggered and could spend three bottles on one long spill. The firmware now takes one sample when an event starts and ends the event only after four clean readings in a row, so four bottles cover four separate events. Replayed on our 30-day synthetic data, the four spills take exactly one	ANALYTICS §2.4; firmware `anomaly_engine.cpp` (commit `64e6289`)

#	Question	Say	Evidence
		bottle each. Written and unit-tested; the bench test is its first run on hardware.	

F. Power and communications

#	Question	Say	Evidence
F1	What's the power budget?	About 4.1 Wh a day nominal including an assumed beacon load (3 W strobe, 2 % duty, 12 h a night), 4.9 with a 20 % margin. The LiFePO4 pack's 61 Wh usable gives about 12 days without sun; without the beacon it would be 15. A 20 W panel gives about 39 Wh on a monsoon day (3 sun-hours, 35 % losses). It's a preliminary estimate: we measure the pump, LTE peaks and beacon on the bench.	backup A3
F2	What if there's no mobile network?	Readings are logged to a microSD card, the light works without any network, and LTE uploads and alerts go out when coverage returns, with SMS as the fallback.	deck slide 5; cost sheet C-03

G. Site, safety and permits

#	Question	Say	Evidence
G1	Will you get permission for a bridge and a river?	We need it from the bridge owner for the brackets, and from the river authority and the pollution board for the pilot. That's why the pilot is planned "with permission" and comes after the tank test.	deck slide 12
G2	Could the light confuse boats?	Good point: red and green are navigation colours in India's buoyage system. We'll ask the waterway authority; it may need to be a yellow or a shielded, bank-facing light.	research §1.4
G3	What about theft or vandalism?	It sits at a pier, not out in the river; the gate pins can take padlocks; and under the service model we replace damaged units. The pilot will show how real this risk is at Kopargaon.	—

H. Business

#	Question	Say	Evidence
H1	Who buys it, and why?	Pollution boards, to catch discharges and know when to sample; water-works, to close an intake in time; irrigation departments; and industrial estates that want an independent downstream record.	02_BUSINESS_SLIDES.md §1
H2	What does it cost?	About ₹1.57 lakh for one float before GST, plus a ₹29k bridge kit (pole and brackets) bought once per site. At 100 units it's about ₹1.24 lakh a float. We priced all 66 part numbers from Indian suppliers; the four probes are 44 % of the float. **And nobody has to buy it:** a board pays ₹12,000 a month for the service, against ₹35–84 lakh for one real-time station. *If cost is pressed:* a Lite float with pH, EC, temperature and level only would be about ₹1.13 lakh. *If asked why it's above our FDC's ₹1.34 lakh:* that sheet was for the lighter v2 layout. The R3 steel structure, fasteners and labour were missing; the probes cost the same.	backup A5; costed BOM (23 Oct)
H3	Isn't a real-time station better?	It measures more parameters with lab-grade instruments; CSE puts the instruments at ₹35–40 lakh. We complement it: many cheap floats between stations for early warning, with a sample that tells the lab where to look.	business §2
H4	How do you make money?	Monitoring as a service: ₹12,000 a month per float on a 5-year contract (the same term as the O&M in station tenders), covering hardware, installation, monthly cleaning and calibration, alerts and data. At the 100-unit cost the hardware and installation (≈ ₹1.63 lakh) are paid back in about 20 months; over	business §1, §4; costed BOM

#	Question	Say	Evidence
		5 years revenue is ₹7.2 lakh, costs about ₹3.86 lakh, margin about 46 %. The ₹77k a year cost to serve is ₹45k of running costs plus the hardware spread over 5 years. Boards that prefer to own it can buy it for about ₹3 lakh installed plus a maintenance contract.	
H5	How big is the market?	CPCB's latest assessment (published 2025, 2022–23 data) counts 296 polluted river stretches and locations on 271 rivers; Maharashtra has 54, still the most. At three floats per counted site that's 162 floats in Maharashtra (≈ ₹2.33 crore a year) and 888 nationally (≈ ₹12.8 crore), before water-works and industry. It's illustrative: the pilot has to identify real deployment sites.	backup A6; PIB 16 Mar 2026
H6	Who are your competitors?	Real-time stations, manual grab sampling, multiparameter sondes, buoys and drone boats, and open loggers like EnviroDIY's Mayfly. We haven't found one that combines a pier-guided float, an on-board decision, a sample and a public light at this cost.	research §5
H7	How will you manufacture it?	One rotomould makes both hulls. The deck and carriage are cut, bent and welded 6061; the pole and brackets are standard 316L pipe and plate; the electronics are off the shelf. Local fabricators can build it, and we assemble and test every float.	business §3
H8	What does maintenance cost?	Mainly monthly visits, lab samples and probe consumables. Our estimate is about ₹77k per float per year: ₹30k of hardware (float and bridge kit) spread over 5 years, ₹18k of monthly visits, ₹14k for four lab samples, ₹9k of consumables and about ₹7k for installation, data and the dashboard. The pilot will measure the real service effort.	costed BOM, "Business model" sheet

I. Team and process

#	Question	Say	Evidence
I1	What changed between IDC and FDC?	The pole moved to the bow (in v2 it ran through the electronics box). The real CAD mass took us from 25.4 to 48.5 kg, so the hulls grew. A sliding collar became 8 rollers, hobby sensors became industrial RS485 probes, and we added the sampler, wiper and light.	deck slide 10
I2	Which 3DEXPERIENCE / SOLIDWORKS tools did you use?	SOLIDWORKS for the 66-part assembly, SOLIDWORKS Simulation for the pole and bracket, Pack and Go for the submission, and ENOVIA Project Planner for our team schedule. Next is Flow Simulation, to confirm the drag.	FDC post; ENOVIA bonus post
I3	What's your biggest risk?	The unknown flood-stage range at the site. We'll get the gauge record, design the travel to it, and include a procedure for removing the float on flood warnings.	C1
I4	Why should you go to the finale?	The design is complete and checked against every requirement, and we're open about what's left. Before the 19–20 Nov finale we'll float our 1:5 scale model and run the decision logic on the ESP32 with a real probe, so there we can show measured results, not just calculations. The full-size tank test comes after that.	deck slide 8, backup A8

J. The hardest questions (our own review, 24 Oct)

#	Question	Say	Evidence
J1	Can you detect oil if conductivity doesn't change?	Not reliably. AquaSentinel is a multi-parameter early-warning system, not a chemical identifier. Oil that changes none of pH, EC, DO or turbidity can be missed. Discharges with ions, acids, alkalis, organic load or sediment are what it targets; the lab sample confirms what it was. An oil or fluorescence sensor is an option if a site needs it.	A4
J2	Why pH above 9 if you claim CPCB Class E?	Those are our operating alarm limits, not one CPCB class. CPCB Class E is pH 6.0–8.5 and EC up to 2,250 µS/cm; DO ≥ 4 mg/L is the Class C/D criterion. Limits are set per site from its designated use and the learned baseline.	A4; CPCB designated-best-use criteria

#	Question	Say	Evidence
J3	What's the complete cost at one site?	One float ₹1,57,492 plus its bridge kit ₹29,218 = ₹1,86,710 ex-GST for the prototype; each float has its own kit at its own bridge. Installation ≈ ₹15,000. At 100 units: ₹1,48,055 + installation. Tooling (one rotomould + jigs, ₹1.25 lakh) is spread over 100 floats, ₹1,250 each.	A5; costed BOM
J4	How does ₹12,000 a month recover the hardware?	Over a 5-year contract: revenue ₹7.2 lakh; hardware + installation ≈ ₹1.63 lakh; running costs ≈ ₹45k a year (₹2.23 lakh); margin ≈ 46 %, payback ≈ 20 months. On a 3-year contract the margin would be ≈ 31 %, so we ask for 5 years or a setup fee.	H4
J5	Has it ever run in a real river?	No. It is designed, calculated and simulated. Next: the 1:5 model in water and a bench demo with a real probe, then a full-size tank test, then a permitted 30-day pilot with lab checks.	slide 9, A8, A9
J6	Does 2.66 include debris impact and anchors?	No. 2.66 is our most conservative static load case: 142 N drag (½·ρ·v^2·Cd·A, Cd 1.05, 0.120 m^2, 1.5 m/s) plus 500 N of debris as a quasi-static force. Dynamic impact, fatigue, vortex shedding, clamp slip, welds and anchors are still to verify, with the bridge owner's engineer.	A2
J7	Is the 12-day autonomy measured?	No, it's a preliminary estimate that now includes an assumed beacon load. We will measure the complete system on the bench.	A3
J8	How do you stop a faulty sensor reporting normal water?	A failed probe read sets a sensor-fault state that is reported, never 'normal water'; the baseline is saved to flash. These are in the firmware; the bench test verifies them.	A4
J9	Is the sample legal evidence?	No. Under the Water Act §21 only the Board's own sampling counts. Ours is a triage sample that tells the Board when and where to take a formal one.	A7
J10	Why not the latest CPCB figures?	We use them: 296 polluted stretches/locations, Maharashtra 54 (assessment published 2025). The older 2022 count was 311 and 55.	slide 2, A6

Don't quote your IDC or FDC grades to the jury. The rules say scores are confidential.

15. Gaps, missing links and claims to verify

Gap	Why it matters	Action
No physical test of anything yet	every performance number is a design value	1:5 float test and bench demo before 19 Nov; record photos and logs
Power budget is not measured; beacon assumed	the 12-day figure could move either way	bench energy test with the full electronics
Detection only on synthetic data	false-alarm rate and sensitivity unknown	replay real data from the bench probe, then the pilot with lab samples
Structure: static case only	dynamic impact, fatigue, anchors, clamp slip, welds unchecked	impact estimate + anchor check; bridge owner's engineer
Value-engineering changes not yet in CAD	6 mm brackets, plate clamps, bought collars, standard anodising	update CAD and re-run the bracket FEA
Firmware alarm limits vs slides	resolved 24 Oct: values match the slides (pH 6-9, DO 4 mg/L, EC 2,250 uS/cm); comments corrected from Class C to operating limits	re-run the firmware unit tests on the board
Costs from listings, not quotes	prices and fabrication hours may differ	two written quotes for probes and fabrication
No customer evidence yet	willingness to pay ₹12,000/month unproven	5 short interviews (MPCB, water-works, WRD, CETP, NABL lab); seek a letter of interest
Flood-stage range at the site unknown	2 m of travel may not be enough	CWC/WRD gauge records for the Kopargaon bridge
Permits	pilot cannot start without them	start applications with the bridge owner, river authority and MPCB
Patent search English-only	a closer Chinese utility model may exist	Chinese-language search before any novelty statement
Git repository layout	tracked files moved into 09_SOURCE_CODE_AND_WEBSITE; a push would break the website	work from a fresh clone; never push from this folder

16. Project folder map and files

Folder	Contents
AquaSentinel_Complete_Project_Dossier.pdf	this document
01_START_PRESENTATION.bat	opens the final deck
01_DESIGN_CAD_AND_DRAWINGS	SOLIDWORKS Pack and Go; assembly STEP/GLB; all 66 part STEP/STL; 6 × A3 drawings; design checks; 2400 px renders; BOM_R3.csv
02_PRESENTATION_SEMIFINAL	AquaSentinel_SemiFinal_FINAL.pptx (and PDF backup), AquaSentinel_Pitch_Script_8min_FINAL.pdf, the 60 s marketing video
03_BUSINESS_COSTING_AND_BOM	costed BOM workbook (every line, source and basis)
04_SIMULATION_AND_ANALYSIS	SOLIDWORKS Simulation reports and plots, detector and plume plots, power budget, design-check data
05_SCALE_MODEL_1to5	1:5 print files (float, stand, parts), previews, print guide
06_ELECTRONICS_AND_FIRMWARE	ESP32 firmware, hardware notes and schematic, bench energy test, bench-demo plan
07_VIDEOS	marketing (60 s), pitch, journey, assembly-and-operation videos
08_SUBMISSIONS_AND_RECORDS	FDC deck and official report, organisers' templates, Product Introduction deck, access email
09_SOURCE_CODE_AND_WEBSITE	full source tree behind the GitHub repository: CAD generator, analytics, website, dashboard, build scripts, prep notes and sources

17. Sources

Cost model sources (S1–S49), looked up 23 Oct 2026:

1. **S1** DFRobot store: SEN0708 RS485 pH sensor $129, SEN0680 RS485 fluorescence DO sensor $199 — https://www.dfrobot.com/search-RS485.html
2. **S2** DFRobot store: SEN0707 RS485 EC sensor K=10 $159 (SEN0706 K=1 $149) — https://www.dfrobot.com/search-conductivity.html
3. **S3** DFRobot store: SEN0710 RS485 turbidity sensor 0-1000 NTU $229 — https://www.dfrobot.com/search-water%20quality%20monitor.html
4. **S4** Evelta (Mumbai): DFRobot SEN0706 Rs 14,414.40 and SEN0709 Rs 17,160 ex-GST, i.e. about Rs 96-97 per USD — https://evelta.com/sensors/liquid-sensors/
5. **S5** Pluang currency page: USD/INR 96.5 (50-day average 95.2) — https://pluang.com/en/tools/currency-converter/usd-inr
6. **S6** IndiaMART: Embsys RS485 optical DO sensor Rs 40,000; RS485 turbidity sensor Rs 38,350 ex-tax (Indian alternatives cost more) — https://m.indiamart.com/proddetail/dissolved-oxygen-sensor-with-rs485-output-for-fresh-water-26441477897.html
7. **S7** Hubtronics: JSN-SR04T-2.0 waterproof ultrasonic Rs 265.25 + GST — https://hubtronics.in/generic/jsn-sr04t-2.0-ultrasonic-sensor
8. **S8** IndiaMART: Espressif ESP32-DevKitC-32E Rs 749 — https://m.indiamart.com/proddetail/espressif-esp32-devkitc-32e-esp32-wroom-32e-development-board-module-for-arduino-2851448645591.html
9. **S9** IndiaMART: SIMCom A7670C LTE Cat-1 module Rs 450-900 (A7670C-LNNV Rs 489); with carrier board Rs 1,900 — https://m.indiamart.com/proddetail/simcom-a7670c-lnnv-lte-cat-1-module-2855267080891.html
10. **S10** Robogears: LiFePO4 12.8 V 6 Ah with BMS Rs 2,200 incl. tax; IndiaMART 12.8 V 6 Ah Rs 660-1,300 — https://robogearsin.odoo.com/shop/10258578-lithium-ferrous-phosphate-lifepo4-battery-of-12-8v-6000mah-with-inbuilt-bms-for-12v-solar-ev-applicat-3642
11. **S11** Loom Solar 20 W 12 V panel Rs 1,299 (Amazon-fulfilled listing, regular Rs 2,500) — https://www.ysvara.com/products/loom-solar-panel-20w-12v-for-home-lighting-system-diy-projects-small-battery-charging7ah-pack-of-1-max-current-1-04-amps-max-voltage-19-25-volts-performance-warranty-25-years
12. **S12** Zbotic: CN3791 12 V MPPT solar charger module Rs 207 + GST — https://zbotic.in/product/cn3791-12v-mppt-solar-charger-module/
13. **S13** Robu: Kamoer KPHM200 12 V peristaltic pump, 260 mL/min, Rs 1,877 incl. GST — https://st.robu.in/product/kamoer-12v-0-7a-10w-260ml-min-kphm200-sw3b16-peristaltic-pump
14. **S14** Robu: 12 V 1/4 in 2-way solenoid valve Rs 596 incl. GST — https://stg.robu.in/?p=975725
15. **S15** Zbotic: RDS3218 20 kg metal-gear waterproof (IP66) servo Rs 1,009 + GST — https://zbotic.in/product/dsservo-rds3218-270-degree-high-torque-digital-steering-gear-with-long-and-short-u4-metal-parts-torque21kg/
16. **S16** Probots: Promax 12 V waterproof strobe light Rs 761.86 + GST — https://probots.co.in/promax-dc-12v-red-strobe-light-flashing-waterproof-silent.html
17. **S17** Robu: 4/5 dBi 3G/4G LTE antenna, SMA, Rs 189 incl. GST — https://aws.robu.in/product/824-960-mhz-1710-2170-mhz-5dbi-gain-dual-band-3g-4g-lte-antenna
18. **S18** IndiaMART: Neubox IP67 polycarbonate enclosure 400 x 350 x 150 mm Rs 1,500 — https://m.indiamart.com/proddetail/rectangular-electrical-junction-box-24692735530.html
19. **S19** Zbotic DS3231 RTC Rs 105 + GST; IndiaMART INA226 module Rs 285; Evelta MAX3485 IC Rs 325-436 ex-GST — https://zbotic.in/product/ds3231-real-time-clock-module-3-3v-5v-precise/
20. **S20** Zbotic: N20 12 V 300 rpm metal gear motor Rs 189 + GST — https://zbotic.in/product/n20-12v-300-rpm-micro-metal-gear-motor/
21. **S21** JLCPCB/PCBWay: 5 x 2-layer 100 x 100 mm boards about $2 plus shipping (Zbotic India cost guide) — https://zbotic.in/how-to-order-pcb-from-jlcpcb-and-pcbway-india-cost-guide/
22. **S22** RS Components: M16 nylon IP68 cable gland about SGD 1.00 per unit — https://sg.rs-online.com/web/p/cable-glands/8229767
23. **S23** Boffin Butler: Tarsons wide-mouth HDPE 250 mL bottles, about Rs 75-96 each incl. GST — https://boffinbutler.com/wide-mouthbottle-hdpe-250ml.html
24. **S24** IndiaMART: SS 316 seamless pipe 2 in Rs 400/kg; 316L SMLS Rs 475/kg — https://m.indiamart.com/impcat/ss-316-seamless-pipe.html
25. **S25** OfBusiness: SS 316/316L HR plate Rs 280-320/kg, CR sheet Rs 300-350/kg — https://www.ofbusiness.com/prices/stainless-steel/316l-ss-hr
26. **S26** IndiaMART: 6061-T6 aluminium plate/sheet Rs 335-415/kg — https://m.indiamart.com/metaaerospace
27. **S27** OfBusiness: LLDPE rotomoulding grade Rs 110-120/kg — https://www.ofbusiness.com/prices/polymers-packaging/lldpe
28. **S28** Aajjo/IndiaMART: rotational moulds, mid-size tank moulds Rs 50,000-72,000; 100 L aluminium lid mould Rs 15,000 — https://www.aajjo.com/product/roto-mould-die-in-ahmedabad-neelam-fabricators
29. **S29** IndiaMART (Techpro, Bengaluru): SS laser cutting Rs 500 per metre — https://m.indiamart.com/techpro-industries/cutting-service.html
30. **S30** Welding cost guide (India): TIG Rs 400-1,000/h machine cost; welder Rs 150-400/h, skilled up to Rs 700/h — https://getcostidea.com/?p=10040
31. **S31** IIT Kanpur workshop rate card (May 2025) lathe Rs 400-500/h, milling Rs 450-550/h; IndiaMART CNC job work Rs 350-550/h — https://www.iitk.ac.in/celp/data/downloadables/Workshop-Job-Card-16-05-2025.docx
32. **S32** IndiaMART: hard anodising Rs 1.50/sq in (A1 Anodizers); anodising Rs 150/sq ft (ENP Techno, Thane) — https://m.indiamart.com/a1-anodizers/anodizing-services.html
33. **S33** IndiaMART: UHMW-PE rod Rs 300-460/kg, UHMW-PE 500 sheet Rs 295/kg; POM/acetal rod Rs 280-450/kg — https://m.indiamart.com/impcat/uhmwpe-rods.html
34. **S34** IndiaMART: rigid PU foam chemicals Rs 170-335/kg — https://m.indiamart.com/impcat/polyurethane-chemical.html
35. **S35** Aajjo (Ambe Plast): 160 mm PE100 PN6 pipe Rs 174/m — https://www.aajjo.com/product/160-mm-underground-hdpe-pipes-in-ahmedabad-ambe-plast
36. **S36** STIC (June 2025): STP/ETP water package incl. BOD, COD, DO, pH, TSS Rs 3,500/sample; DPCC BOD Rs 1,050, COD Rs 620 — https://sticindia.com/resources/Analysis%20Charges%20-%20from%2001-06-2025.pdf
37. **S37** The Week (May 2025): Delhi DPCC 32 real-time water stations for about Rs 22 crore; Aug 2025 tender raised to 41 with 5-year O&M — https://www.theweek.in/wire-updates/national/2025/05/15/des48-dl-yamuna-water-monitring.html
38. **S38** CSE: CPCB real-time station instruments Rs 35-40 lakh; Lok Sabha reply 5115: Rs 50-84 lakh per station with systems — https://www.cseindia.org/cpcbs-real-time-water-quality-monitoring-4587
39. **S39** CPCB 2022: 311 polluted river stretches on 279 rivers; Maharashtra 55, the most (Lok Sabha reply AU2546) — https://eparlib.nic.in/bitstream/123456789/1467192/1/AU2546.pdf

40. **S40** CPCB reply to NGT (OA 413 of 2025): about 4,922 monitoring locations, 2,265 on rivers, mostly manual sampling — https://www.greentribunal.gov.in/sites/default/files/news_updates/RESPONSE%20BY%20R-5%20CPCB%20IN%20OA%20NO.%20413%20of%202025%20ABHISHT%20KUSUM%20GUPTA%20VS%20UNION%20OF%20INDIA%20AND%20ORS..pdf
41. **S41** IMARC: India water analytical instruments market $150.4 M (2024), $253.4 M by 2033, 6.0 % CAGR — https://www.imarcgroup.com/india-water-analytical-instruments-market
42. **S42** MarketsandMarkets: India environmental monitoring $428.5 M (2025), $616 M by 2030, 7.5 % CAGR — https://www.marketsandmarkets.com/Market-Reports/geography/environmental-monitoring-market/India
43. **S43** YSI: EXO2 multiparameter sonde $7,090-8,770, sensors sold separately — https://www.fondriest.com/ysi-exo2-multi-parameter-water-quality-sonde.htm
44. **S44** Libelium Smart Water Xtreme: RS485/SDI-12 IP68 probes, price on request — https://www.libelium.com/libelium-pushes-the-water-quality-market-ahead-with-its-new-smart-water-xtreme-monitoring-platform/
45. **S45** Eruvaka (India): floating buoys for aquaculture ponds (DO, pH, temperature) — https://yourstory.com/2014/01/fishing-right-waters-eruvaka-technologies
46. **S46** Kisters case study: CPCB data validation for 40 real-time stations on the Ganga (2022-2027) — https://www.kisters.eu/success-story/improving-water-quality-monitor/
47. **S47** India Science & Technology portal: NIDHI-PRAYAS prototype grant up to Rs 10 lakh per innovator — https://www.indiascienceandtechnology.gov.in/hi/funding-opportunities/startups/nidhi-promoting-and-accelerating-young-and-aspiring-innovators
48. **S48** IndiaMART: Modbus RS485 pH controllers Rs 8,500-12,000; RS485 conductivity sensors Rs 7,500-20,499 — https://m.indiamart.com/impcat/conductivity-sensors.html
49. **S49** Previous team cost sheet (v2 layout): hardware/AquaSentinel_Cost_Estimate_BOM.csv, lines sum to Rs 1,38,000 incl. GST — hardware/AquaSentinel_Cost_Estimate_BOM.csv

Prior art (24 Oct 2026): CPCB WQM overview (cpcb.nic.in/ngrba/WQM_overvew.php); CSE; YSI EXO2 (ysi.com/EXO2); NexSens CB-450 (nexsens.com); Teledyne ISCO 6712 via RS Hydro; Google Patents CN117446093B, US8352213. Market: PIB 16 Mar 2026 (CPCB polluted river stretches); IMARC.